CHAPTER XXXIII

ON LINEARITY

The conception of linear transformation thus plays the same part in affine geometry as congruence plays in general geometry; hence its fundamental importance. (547) HERMANN WEYL

It is instructive to compare the mathematical apparatus of quantum theory with that of the theory of relativity. In both cases there is an application of the theory of linear algebras. (215) W. HEISENBERG

This "perturbation theory" is the complete counterpart of that of classical mechanics, except that it is simpler because in undulatory mechanics we are always in the domain of *linear* relations. (466) E. SCHRÖDINGER

As a result of experimental research on association, in 1904, I was led to show the complexity of the factors governing evocation. . . . And I have often insisted since then on this essential idea, in opposition to the simple schema of linear associative connection. (411) HENRI PIÉRON

We have already had several occasions to mention the 'plus' or additive issues as connected with linearity. This problem is of structural and linguistic as well as empirical and psycho-logical semantic importance. It is sufficient for our purpose at present that we should notice two facts; namely, (1) That in one dimension, linearity expresses the *relation* of proportionality; (2) That the problems of linearity are dependent on the *relation* of additivity.

The structural notion of additivity is of great antiquity. Being the simplest of such notions, it naturally originated very early in our history. The earliest records show that the Babylonians and the Egyptians used the additive principle in their notation. Our primitive ancestors, long before any records were written, had similar structural conditions present, open for investigation and reflection, that we have today. That this was the case is not a mere guess. Otherwise we would still be at their stage of development. Some beginning had to be made somewhere. There is little doubt that the men of remote antiquity presented many types of make-up, as we do today. Some, for instance, were more curious than others; some more inventive, some more reflective . , which, as we know today, is found even among animals. These more gifted individuals were, as usual, the inventors, discoverers, and builders of systems and language of their period. They could not long fail to recognize the fact that a stone *and* a stone, or a fruit *and* a fruit are *different* from *one* stone or *one* fruit. For instance, the two stones might have saved the early observer's life in defence, or the two fruits might have satisfied his hunger or thirst, where one would not have done so. An accumulation of objects was obviously somehow different from a single object. As these problems were often of vital importance to their lives, names for such accumulations of objects began to be invented, and one and one was called two, two and one was called three , . Number and mathematics were born as a structural semantic life-necessity

for a time-binding class of life. They were an expression of the neurological structure and function and of the tendency toward induction.

In the beginning, names and generalizations were made from the simplest brute facts of life, and our primitive ancestors did not realize, that these crude generalizations might not have a structural validity, which they seldom doubted that they possessed, even as we today, seldom doubt. Those primitive scientists, (and we today differ very little from them), having produced *terms*, objectified them, and began to speculate about them. Let us examine some examples of such primitive mathematical speculations. Addition, of course, by which we *generate numbers*,—one and one make two, two and one make three . , was all-important. They could not miss the simple fact that three, which is equal to two and one, by definition, is more than two or one. A primitive generalization; namely, that the sum is always more than the summands taken separately, was still further generalized to a postulate that a part is smaller than the whole. This generalization has hampered mathematics almost up to our own day, and for many thousands of years it prevented the discovery of the notion of mathematical infinity, which we have already discussed in Chapter XIV.

It must be noticed that such generalizations involve *s.r*, which are objective and un-speakable. If verbally formulated they should have a structure similar to that of the facts, otherwise they are fanciful and vicious, because not properly formulated. When formulated they become public structural facts (*s.r* are personal, individual, non-transmittable, and un-speakable) and so they may be criticized, improved, revised, rejected , . All human history shows that the correct structural formulation of a problem is usually as good as the solution of it, because sooner or later a solution always follows a formulation.

After many thousands of years—in fact, practically only the other day—it was found that these primitive generalizations were in general not valid. Negative numbers were invented, and two plus minus-one was no more three but one, $2+(-1)=1$. The sum was *no* longer greater than its summands. The usual tragedy takes place here also. A few people know the facts, but the old primitive structural *s.r remain* in some of these few, as well as in the great majority of us who did not even know the facts. That such structural *s.r* do not vanish quickly, or generally, is proven again and again throughout history. We see it very clearly in the problems of 'infinity', or $\bar{E}$ geometries, or $\bar{N}$ physics. But the most pathetic sight is to see scientists who have *rationalized* the technique without a deeper re-education of their *s.r*. This is most clearly seen in the case of many writers on the foundations of mathematics, the Einstein theory or on the newer quantum mechanics. They *feel* in the old structural way, they *rationalize* in the new, hence their works are full of self-contradictions. Readers and students alike feel how 'difficult' and messy the whole subject is. As a matter of fact, the new theories are neither messy nor difficult. They are really much simpler and easier than the old theories, *provided* our structural *s.r* are purged of the primitive structural tendencies to which every one of us is heir. When this semantic re-education of our structural feelings is accomplished

it is the old that becomes 'unthinkable' and incomprehensible, because it gives such a structural mess.

Something similar might be said about a feeling deep-rooted in all of us namely, the 'plus' feeling. In all the advances of science we struggle against it. For instance, the example of the green man-made leaf given previously shows clearly that man-made affairs may with some plausibility be considered as 'plus' affairs, but not so with non-man-made natural leaves, which appear not as 'plus', but functional affairs, where the greenness was structurally *not added* but happened, or became. As a structural fact, the world around us is *not* a 'plus' affair, and requires a functional representation. In chemistry, for instance, does hydrogen 'plus' oxygen produce water, H_2O? If we mix the two gases, two parts of hydrogen with one of oxygen we do not get water. We must first pass a spark through the mixture, when an explosion occurs and the result becomes *water*, a *new* compound quite *different* from its elements or from a mere mixture of them. Does one gallon of water and one gallon of alcohol make two gallons of a mixture? No, it makes less than two gallons. Does light added to light make more light? Not always. The phenomena of interference show clearly that light 'added' to light sometimes makes darkness. Four atoms of hydrogen, of atomic weight 1.008, produce, under proper conditions, one atom of helium, not of atomic weight 4.032, but of atomic weight 4. The 0.032 has somehow mysteriously vanished. Such examples could be quoted endlessly. They show unmistakably that structurally this world is not a 'plus' affair, but that *other* than additive principles must be looked for.

The struggle against this 'plus' feeling is quite evident, but often unsuccessful, in scientific literature. Man 'is' an animal 'plus' something. Life 'is' 'dead matter', 'plus' some 'vitalizing principle',. In scientific literature we find curious expressions: as for instance, 'It is impossible to express the conduct of a whole animal as the algebraic sum of the reflexes of its isolated segments'; or, 'The individual represents heredity *plus* environment'; or, 'That the abstraction does not merely take away from a number of engram groups some components and combine the rest into one sum, but forms thereby a new psychic structure is self evident and is in no way peculiar to the psyche. Thus a clock work is as little the mere *sum* of its little wheels as a human being is the *sum* of his cells and molecules'; and later on, 'to be exact the ego consists of the engrams of all our experiences *plus* the actual psychism'. There is endless material that might be quoted, but for our purpose these few samples will suffice. We do not give them with the purpose of citing authoritative examples of the need of non-plus considerations. Far from it. We do it to emphasize the astounding fact that, although the best men in their fields have vaguely felt this necessity, yet even they become a prey to this very old structural linguistic semantic tendency. In all three cases quoted the authors were of the best we have. They have fought all their lives against the 'plus' tendency and methods; and yet, if they succeed in eliminating this tendency from one part of their subject, they plant it quite obviously somewhere else. We see that

we are dealing here with an ingrained psycho-logical tendency which can be remedied only by a fundamental, $\bar{A}$, structural, semantic investigation.

Let us analyse these quotations. In the second case, we hear, after a successful attack on *plus* tendencies, a statement that the 'individual represents heredity *plus* environment'. Is this statement true? Let us take examples. There are certain fishes which are heliotropic and swim toward the light, but if we change the temperature of the water they become negatively heliotropic and swim *away* from the light. Is this most complex activity of the organism-as-a-whole a 'plus environment' fact, or does the change of temperature produce some fundamental functional changes? When, for instance, a good mother rat, having been put on a different though still abundant diet, which is deprived of some minute amount of special vitamins, begins to eat her litters, is this again a 'plus' reaction, or is it a most complex functional change of the organism-as-a-whole? Or when a human being, because he received in childhood an 'emotional' shock through outside events (action or language of parents, for instance) develops a functional disorder, or even a physical ailment, is this again a 'plus environment' problem? Or, when chickens fed on eggs laid by hens kept without sunlight or violet rays, or which have only received sunlight through a glass window, develop rickets and soon die, though they do not do so when the glass windows are removed and the sunlight is allowed to operate directly upon the hens. Is this again a 'plus environment' example?

One 'Smith' and one 'Smith' make two 'Smiths', as far as theatre or railway tickets are concerned, but in life, under proper conditions, they form a family and very often many more than two 'Smiths' come out of such 'addition'. How about their work? Is it a mere sum? In the case of inventors who may have been influenced by one or many men directly or indirectly, do their inventions produce a sum of the work of as many men? Surely the steam engine or the dynamo produces more work than not only the inventors, but the series of other men who have been indirectly responsible for the inspiration of the inventors, could ever have produced. So again it is not a 'plus' affair.

In the third case we see the author attacking the 'plus' tendency on one page, and planting another 'plus' a few pages further on, which implies at once some objectified *additional* entity. In this respect it must be noticed that this *additive* tendency represents a partial and important structural and semantic *mechanism of identification*, and to deal successfully with it, we must clear up the problem connected with the *additive* tendency.

The numberless and endless 'philosophical' volumes, for instance, which have been written about the 'body-soul' problems, show the tremendous structural and semantic importance of the clearing up of this 'plus' versus 'non-plus' issue. The reader may recall that the $\bar{A}$, the $\bar{E}$, and the $\bar{N}$ systems have one underlying structural metaphysics. The $\bar{E}$ systems deal with non-linear equations and with curved lines, of which the linear equation and the straight line, (one of zero curvature), are only particular cases. And the general theory of Einstein, which is the foundation of $\bar{N}$ systems, also introduces non-linear equations. Ought we to be surprised to find that a $\bar{A}$-system must also solve

this difficult structural and semantic problem of linearity versus non-linearity, of additivity versus non-additivity?

Indeed the problems demanding our attention are extremely baffling and difficult. Even in such a perfected science as physics, we have great difficulties in using non-linear equations, and are still at the stage where we solve few equations other than linear ones. To make any progress at all we must start with the *simplest* available problems in this field; namely, mathematical problems. The main point at this stage is not a solution of the problem but its formulation. When formulated and brought to the attention of mankind, there is no doubt that it will be eventually solved.

To better understand the additive principle, let us consider a group of elements, the individuals of which we denote by letters a, b, c, d , . Let us take two or more of these elements and produce a synthesis which results in a third or n-th entity. Let this synthesis be of such a nature that the characteristics ascribed to the elements are also present in the resultant synthesis, in other words, let them have the so-called group characteristic. If our elements are, for instance, numbers, the new synthesis is also a number and belongs to the original group. We must notice that the problem of *order* is important in the formulation of the additive principle. If a and b are the two elements the synthesis of which we define, we must be clear that a first and b second, or b first and a second, must be recognized in the synthesis. Let us assume also that only the two alternative orders a and b, or b and a, are of importance in this case. The commutative law asserts that a plus b is equal to b plus a, $a+b=b+a$, which means that the two possible alternative orders give equivalent results. We must notice that this does not mean that order does not enter into this synthesis; in such a case the above mentioned commutative law would make no assertion at all. It is of importance that order should be involved in the synthesis. It is a matter of indifference only as far as equivalence by a commutative law is concerned.

We should notice for our purpose that the synthesis has the 'same' characteristics as the elements had. In other words, if we know the characteristics of the elements we know the characteristics of the result. For instance, if the elements were numbers, the result will be a number, and no characteristic absent in the elements will appear in the result. This predictability from the characteristics of the elements to those of the result is perhaps one of the most striking characteristics of additivity. On the one hand, it allows us to foretell the future; on the other hand, it limits considerably the applicability of the additive principle. It is obvious that when we combine elements, and the results have *new* characteristics absent in the original elements, the new problems are structurally no more of an additive character, and the synthesis must be different.

Only a few of the simplest entities in physics possess additive characteristics. If we take, for instance, 'weight' or 'length' or 'time', we see that these units are additive. One pound, or inch, or second, if added respectively to one pound, or inch, or second, gives us two pounds, or two inches, or two seconds.

Not so, however, with temperature, or density, or many other derived magnitudes, as we call them. If we have a body of temperature of one degree and combine it with another body of equal temperature the synthesis will not have a temperature of two degrees, (as in the case of weight), but of one degree. This applies to density . , two bodies of density one each will not give us a body of density two, but of density one.

Before further analysis of the problems of linearity and additivity, it will be well to consider a few definitions.

If an entity u is changed into an entity v by some process, the change may be regarded as the result of an operation performed upon u, the operand, which has converted it into v. If we denote the operation by f, then the result might be written as $v = fu$. The symbol of the operation f is called the *operator*. We are familiar with many such; indeed the symbols for all mathematical operations may be treated as operators. So for instance the symbol $\sqrt{\ }$ indicates the operation of extracting the square root. If we deal with a range of values for a variable x, what we have defined as the function symbol $f(x)$ may be treated as an operator whose operation on x may be indicated by the symbol fx. The operation of differentiation may be symbolized by D, the result of whose operation on the variable u, Du is the derivative of u. The sign of the definite integral $\int_b^a$ may be taken as indicating an operation which converts a function into a number , .

It is important to know that many of the rules of algebra and arithmetic when defined in this way, give rise to a calculus of operations. The fundamental notion in such a calculus is that of a product. If u is operated upon by f the result v is indicated by fu, or symbolically, $v = fu$. If v in turn is operated upon by g the result w is indicated by gv, or symbolically, $w = gv = gfu$, whence the operation gf which converts u directly into w is called the product of f and g. If this operation is repeated several times in succession the usual notation of powers is used, for instance $ff = f^2$, $fff = f^3$, . Not applying the operator at all, which we would denote by f^0, leaves u unchanged, which we indicate symbolically by the equation $f^0u = u$. The operator f^0 is equivalent to multiplication by 1, $f^0 = 1$, whence f^0 may be called the *idem operator*. We see also that the law of indices holds; namely, the $f^m \times f^n = f^{m+n}$.

For our purpose we will analyse only one special case; namely, where we have u, v and $u+v$ as operands, and such an operator, f, that $f(u+v) = fu + fv$. Expressed in words, this means that the operator applied to the sum of the two operands gives a result equal to the sum of the results of operating upon each operand separately. Such a special operator is called a *linear*, or distributive, operator.

In terms of functions we would have $f(x+y) = f(x) + f(y)$ which may be called a functional equation. It has been proved that such a functional equation has one type of solutions; namely, when f is equivalent to a multiplication by a *constant*, or $fx = cx$. This fact is of great importance for us. Many problems in science are stated in terms of variation. For purposes of analysis a statement

that 'x varies as y' is written $y=kx$, where k is called a factor of proportionality, which enables us to convert a statement of variation into an equation. If y varies inversely as x, we write $y=k(1/x)$ or $y=k/x$. A multiplication by a *constant* thus introduces a relation of proportionality, hence the importance of proportionality in a world where constants are present.

It must also be noticed that the two fundamental operations of the calculus are *linear* without being equivalent to multiplication by a constant. These are: 'the derivative of the sum is the sum of the derivatives', that is, $D(u+v)=Du+Dv$; and 'the integral of the sum is the sum of the integrals', that is $\int(u+v)dx=\int udx+\int vdx$. But as the fundamental notion of the calculus is to substitute for a given function a *linear* function, in other words, to deal with curves as the limits of vanishingly small straight lines, this linearity underlies structurally all fundamental assumptions of the calculus, and one might say with Weyl that 'one here uses the exceedingly fruitful mathematical device of making a problem *linear* by reverting to infinitely small quantities'.[1]

A vector is defined roughly as a line-segment which has a definite direction and magnitude, and any quantity which can be represented by such a segment is defined as a vector quantity.

The addition of vectors is defined by the law of the parallelogram, as in the case of two forces. It should be noticed that because of this definition the sum of two vectors *differs* in general from the arithmetical sum of the lengths, and only collinear, or parallel vectors obey the arithmetical summation law.

The introduction by definition of mathematical entities which obey different laws from the usual arithmetical laws is an important structural and methodological innovation. It gives us the useful *precedent* of defining our *operations* to suit our needs. The vector calculus accepted as the definition of the sum of two vectors the law established *experimentally* in physics for the sum of two forces; and so the vector calculus from the beginning was structurally a particularly useful language in physics. Only since Einstein has the value and importance of the vector calculus for physics become generally appreciated.

If we have two vectors, **a** and **b**, starting from a common origin O and complete the parallelogram as in Fig. 1, then the diagonal of the parallelogram will be the required sum, $\mathbf{a}+\mathbf{b}$, by *definition*.

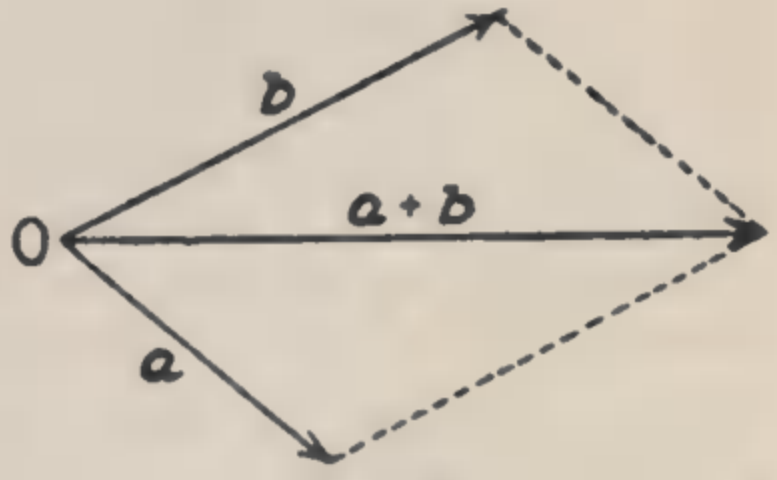

FIG. 1

If we choose two co-initial vectors of unit length, one on the X axis, and the other on the Y axis, and call them **i** and **j**, we can always represent any vector **x** as the sum of two vectors, one of which is the projection of **x** on the X axis, and the other the projection of **x** on the Y axis. (See Fig. 2.) Let us call these vectors $\mathbf{x}'$ and $\mathbf{x}''$ respectively. Then $\mathbf{x}=\mathbf{x}'+\mathbf{x}''$, by definition. But $\mathbf{x}'$ differs from **i** in length only, hence it can be obtained by multiplying **i** by

an appropriate number, say a. Similarly, $\mathbf{x}''$ can be obtained from $\mathbf{j}$ by multiplying $\mathbf{j}$ by b, and so, in symbols, $\mathbf{x}'=a\mathbf{i}$, $\mathbf{x}''=b\mathbf{j}$, and $\mathbf{x}=a\mathbf{i}+b\mathbf{j}$. All vectors of the plane can be obtained from $\mathbf{i}$ and $\mathbf{j}$ in this form. The numbers a and b are called *components of* $\mathbf{x}$.

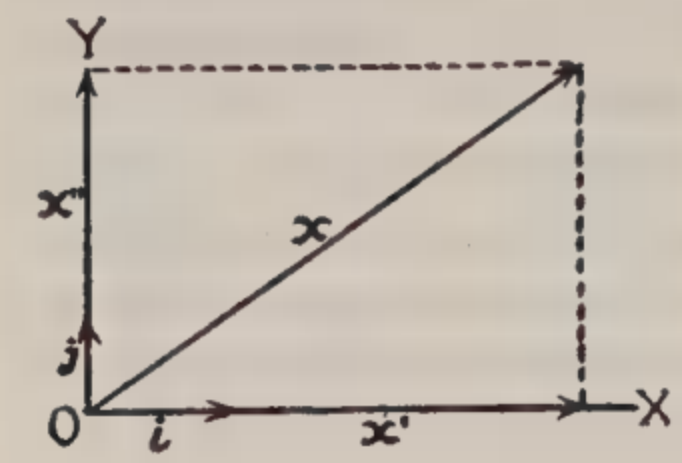

FIG. 2

Now that we know how to express a vector in terms of its components; namely, $\mathbf{x}=a\mathbf{i}+b\mathbf{j}$, let us consider a vector function $f(\mathbf{x})$ which satisfies the equation $f(\mathbf{x}+\mathbf{y})=f(\mathbf{x})+f(\mathbf{y})$. We may take $a\mathbf{i}=\mathbf{x}$ and $b\mathbf{j}=\mathbf{y}$ and $\mathbf{x}+\mathbf{y}=\mathbf{z}$ then we have $f(\mathbf{z})=f(\mathbf{x}+\mathbf{y})=f(a\mathbf{i})+f(b\mathbf{j})$. But since a and b are numbers, we have $f(a\mathbf{i})=af(\mathbf{i})$, and likewise, $f(b\mathbf{j})=bf(\mathbf{j})$; so that $f(\mathbf{z})=af(\mathbf{i})+bf(\mathbf{j})$. But $f(\mathbf{i})$ is itself a vector and therefore expressible in the form $a'\mathbf{i}+b'\mathbf{j}$, and, $f(\mathbf{j})=c\mathbf{i}+d\mathbf{j}$. Hence, $f(\mathbf{z})=a(a'\mathbf{i}+b'\mathbf{j})+b(c\mathbf{i}+d\mathbf{j})=(aa'+bc)\mathbf{i}+(ab'+bd)\mathbf{j}$. In general, the components are the coefficients accompanying $\mathbf{i}$ and $\mathbf{j}$, and so we have the components of $f(\mathbf{z})=f(\mathbf{x}+\mathbf{y})$ in terms of the components of $\mathbf{z}$; and we see how the components of a vector are changed into the components of the linear vector function of the vector.

In general terms, a continuous vector function of a vector is said to be a *linear* vector function when the function of the sum of any two vectors is the sum of the functions of those vectors; that is, the function f is linear if $f(\mathbf{r}_1+\mathbf{r}_2)=f(\mathbf{r}_1)+f(\mathbf{r}_2)$, whence, if a be any positive or negative *number* and if f be a *linear* function then the function of a times $\mathbf{r}$ is a times the function of $\mathbf{r}$; $f(a\mathbf{r})=af(\mathbf{r})$.

Linear vector operators are also defined by a similar equation; namely, $L(\mathbf{a}+\mathbf{b})=L\mathbf{a}+L\mathbf{b}$.

Let us recapitulate. If we take the functional equation $f(x+y)=f(x)+f(y)$, which might be used as a definition of *linearity*, and which is based on *additivity*, and take $x=y=1$, then we have $f(1+1)=f(2)$ and also $f(1)+f(1)=2f(1)$; and so our original equation becomes by substitution $f(2)=2f(1)$. It is obvious that the original equation, $f(x+y)=f(x)+f(y)$, is the source of indefinitely many such relations for particular numbers. For instance, $f(3)=f(2+1)=f(2)+f(1)$; but, in accordance with what we obtained before, $f(2)=2f(1)$; so that $f(3)=2f(1)+f(1)=3f(1)$, and in general, $f(x)=xf(1)$. So, if we have an equation $f(x+y)=f(x)+f(y)$ for numbers, we know that we can obtain the value of this function for any x if we know it for 1. If we denote the function of 1, which is a *constant*, by $f(1)=k$, we have the general form of the function which satisfied $f(x+y)=f(x)+f(y)$ expressed by $f(x)=kx$. In words, a functional equation of the above type; namely, a function of the sum equal to the sum of the functions, has only one possible type of solution; namely, when f is equivalent to a multiplication by a constant, or, $f(x)=kx$. But this last means proportionality. The values of the function are proportional to the arguments (variables). In fact, let us consider two arguments, that is, two values of the independent variables x and y. We have, as shown before, $f(x)=kx$ and

$(y)=ky$. Dividing the first by the second, we obtain $f(x)/f(y)=kx/ky=x/y$ or, in another form, $f(x)/x=f(y)/y=k$.

Let us consider an example. We know from elementary geometry that if we take an angle α and draw parallels which cut the sides at AA', BB', CC', DD' . , the corresponding intercepts are proportional. In general, the lengths of the segments on the left side are not equal, neither are they equal to the segments on the right side. If we designate the segment AB as x and BC as y, the corresponding segments $A'B'$ and $B'C'$ we may designate as $f(x)$ and $f(y)$, respectively, which means function of x and function of y. $A'B'=f(x)$, $B'C'=f(y)$. But the above intercepts are proportional, which means that $\frac{AB}{A'B'}=\frac{BC}{B'C'}=\frac{AB+BC}{A'B'+B'C'}$.

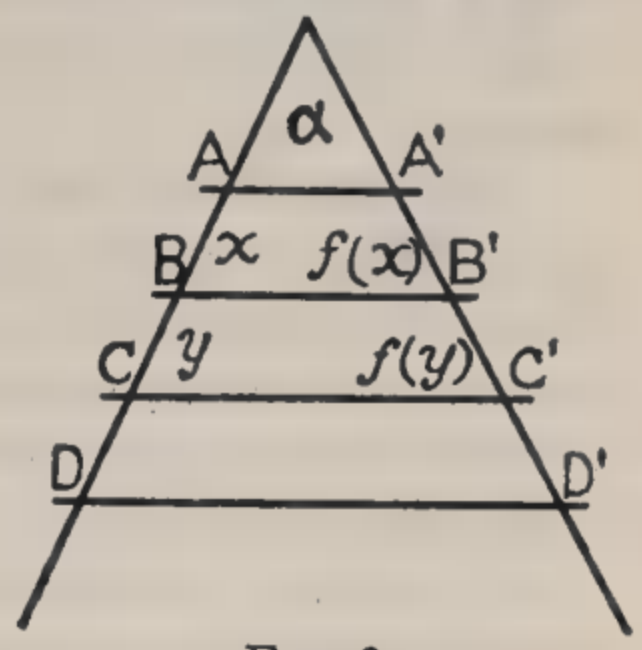

FIG. 3

We easily see from Fig. 3, that $AB+BC=AC=x+y$; and $A'B'+B'C'=A'C'$ and so $A'C'$ on the one hand is $f(AC)=f(x+y)$ and on the other hand it is $f(x)+f(y)$ and therefore $f(x+y)=f(x)+f(y)$. We could multiply examples by taking relations between central angles in a circle and arcs of its circumference. In fact, any problem of *measure* in E geometry could be used as an example.

In our development we started with definite *additive* natural tendencies, not only in our highest, yet undeveloped, mathematics, which we call our daily and scientific language, but also in our lowest, but perfected, language which we call mathematics. In this perfected language the notion of *additivity* is connected with *linearity*, and the *methods* of *approximation* are also founded on additivity and linearity.

Yet the world around us in its more fundamental structural aspects is not additive; and for *adjustment* we must find means of passing from additive tendencies and formulations to non-additive tendencies and formulations. Modern mathematics has developed these methods, and modern physics is beginning to apply them. Let us repeat: the importance of linear functions implies the importance of 'straight' lines. They are important on two counts: first, because they are simpler than all other curves, so that naturally we want to study them before we study other curves, such as, for instance, circles or the other conic sections in elementary geometry; and secondly, because all curves can be approximated by straight lines. This point is very important, as approximation is the most powerful method we have of handling complicated situations.

There are two methods of approximating a curve in the vicinity of a point. If we are interested in the immediate vicinity of a point we approximate the curve by its tangent, as the tangent approximates the curve in the vicinity of a point better than any other straight line. If we want to decrease the error which we make in this approximation, we have only to decrease the vicinity in which we consider it. If we do not want to restrict ourselves to a small

neighbourhood we have to use more complicated methods of approximation. We inscribe into the curve a broken line which consists of segments of straight lines. The beginnings of the study of curves consist in reducing the study of curves to: (1) The study of straight lines connected with the curves' tangents, which is the point of departure of the differential calculus; and (2) The study of the inscribed broken lines, which is the point of departure of the integral calculus.

Now curves represent only the simplest dependences. In other cases we have more complex kinds of functions; for instance, vector functions; but in every case we have *linear* functions, the simplest of their type; and other functions are studied by approximating them in one way or another by linear functions. In using the term 'function', we mean not merely numerical functions but also *operators*, which are to the ordinary functions what ordinary functions are to numbers. A general definition of linearity can be connected with that of proportionality in the following manner. If two variables are proportional, one to another, then to the sum of any values of the first corresponds the sum of the corresponding values of the second.

The simplest part of any field is the consideration of linear, additive questions; linear equations (equations of first degree in algebra), linear differential equations, linear integral equations, linear matrices, linear operators , . But sooner or later we come to the more difficult and more interesting non-linear problems. Perhaps the main importance of the General Theory of Einstein lies in the fact that the equations of physics become *non-linear*. Now, although non-linear equations can be approximated by linear equations, the character of a world determined by non-linear equations must be entirely different from a world determined by linear equations. In a linear world electrons would not repel each other but would travel independently of each other, and there could be no relation between the charges of different electrons. But we know that electrons do repel each other, and attract protons, and that their charges are equal. In physics, if a system can be described by linear differential equations, the causal trains started by different events propagate themselves *without interference*, with simple *addition* of effects.

The properties of systems which can be described by linear differential equations have, as we have already seen, the property of *additivity*. This means that the result of the effects of a number of elements is the sum of the effects separately, and no new effects will appear in the aggregate which were not present in the elements. In such a universe there is 'continuity', fields are superposable, wave disturbances are additive, 'energy' and 'mass' are indestructible , . In such a universe we can have two-valued *causality*, as causal trains started by different events propagate themselves without *interference*, and with simple addition of effects, and the present can be analysed backwards into the sum of elementary events, that is, a two-valued causal analysis is possible.

If our equations are not linear, the effects are not additive and a two-valued causal analysis is not possible.

The joint effect of *two* causes working together is *not* the *sum* of their effects separately[2], and we need ∞-valued causality.

Analytically, if we have *linear* differential equations and we have one solution $y_1 = f(x)$ and another solution $y_2 = F(x)$ then their sum is also a solution; namely, $y_3 = f(x) + F(x)$. If the differential equations are non-linear and if $y_1 = f(x)$ and $y_2 = F(x)$ are two solutions, then $f(x) + F(x)$ is *not* a solution.

Linear problems and linear equations play a very important structural role in science and there is little doubt that linear equations preponderate enormously, although many fundamental events cannot be described by such equations. A universe which can be described by linear differential equations of the second order has definite structural characteristics—in the main in rough accord with observation. As such differential equations give us the tendency of a process, we may use them to describe large-scale phenomena by integration, or the statistical phenomena of great numbers.

Unfortunately, the study of non-linear problems is structurally very difficult and largely a problem of the future.

There is one very important point which we should not miss. We know already that there is a fundamental difference between different orders of abstractions. Physical abstractions have always characteristics left out, and our higher order abstractions are further removed from life, but they have all characteristics included. The problem of sanity being a problem of adjustment, we must somehow correlate these abstractions in which characteristics are *left out* with those which include all characteristics, and so *must* proceed *by approximations*. Mathematical methods, particularly those of the differential and integral calculus, have evolved the best technique of *approximation* in existence today, which, as we have seen, is strictly connected with *linearity* or *additivity*.

A similar urge which prompted us in the expression of our additive tendencies and methods in the structure of language, has led to the production of the calculus. For organisms which abstract in so numerous and such different orders, the methods of the calculus are therefore fundamental psycho-logical devices, conditioning sanity.

In conclusion, we should notice two quite important facts. One of these is that the nervous system, being in a state of nervous tension, cannot structurally be a simple additive affair in all its functions, a fact which every one of us has experienced. Too many stimulations dull, or abolish, or change reaction in an enormous variety of ways. Piéron, as a result of experimenting in association, has not only shown the complexity of these processes, but also reaches the conclusion that the associative connections are non-linear.[3] The other most important point is that structurally the term 'and' implies addition. When we confuse orders of abstractions or levels of analysis, the 'and' additive implications falsify the issues. Thus for instance two atoms of hydrogen *and* one atom of oxygen *and* a spark produce water. The second 'and', at least, is used illegitimately, as it applies to an entirely different level (the spark) from that of the atoms. Linguistically we introduced additive implications,

while empirically we are dealing with most complex non-additive, non-linear higher-degree functions. When we confuse orders of abstractions, as we all do, the 'and' is bound to introduce structurally false implications, which it is very difficult to avoid—the more so since these semantic problems are generally entirely neglected.

CHAPTER XXXIV

ON GEOMETRY

At the same time it will not be forgotten that the physical reality of geometry can not be put in evidence with full clarity unless there is an abstract theory also. . . . Thus, for example, while the term electron may have more than one physical meaning, it is by no means such a protean object as a point or a triangle. (259) OSWALD VEBLEN

Euclidean space is simply a group. (417) HENRI POINCARÉ

It is only in Euclidean "gravitationless" geometry that integrability obtains. (551) HERMANN WEYL

The fundamental fact of Euclidean geometry is that the square of the distance between two points is a quadratic form of the relative co-ordinates of the two points (*Pythagoras' Theorem*). *But if we look upon this law as being strictly valid only for the case when these two points are infinitely near, we enter the domain of Riemann's geometry.* (547) HERMANN WEYL

. . . *parallel displacement of a vector must leave unchanged the distance which it determines. Thus, the principle of transference of distances or lengths* which is the basis of metrical geometry, carries with it *a principle of transference of direction;* in other words, **an affine relationship is inherent in metrical space.** (547) HERMANN WEYL

But before dealing with the brain, it is well to distinguish a second characteristic of nervous organization which renders it an organization in levels. (411) HENRY PIÉRON

Section A. Introductory.

The main metrical rule in geometry is the familiar pythagorean theorem. In 1933 this rule is no longer considered as generally valid outside of the euclidean system, as its proof depends on the doubtful postulate of parallels. It is considered as an empirical generalization in which the relative error decreases when the distances become smaller. Indeed the small element of length, ds, given by the pythagorean rule is considered convenient and reliable in our exploration of the world.

The pythagorean rule states that in any right triangle, ABC, the square of the side opposite the right angle (the hypotenuse) is equal to the sum of the squares of the two other sides (the legs). In symbols, $AB^2 = AC^2 + BC^2$. If we build squares on all three sides of the triangle ABC and denote the areas of the squares by C', A', and B' then we have $C' = A' + B'$.

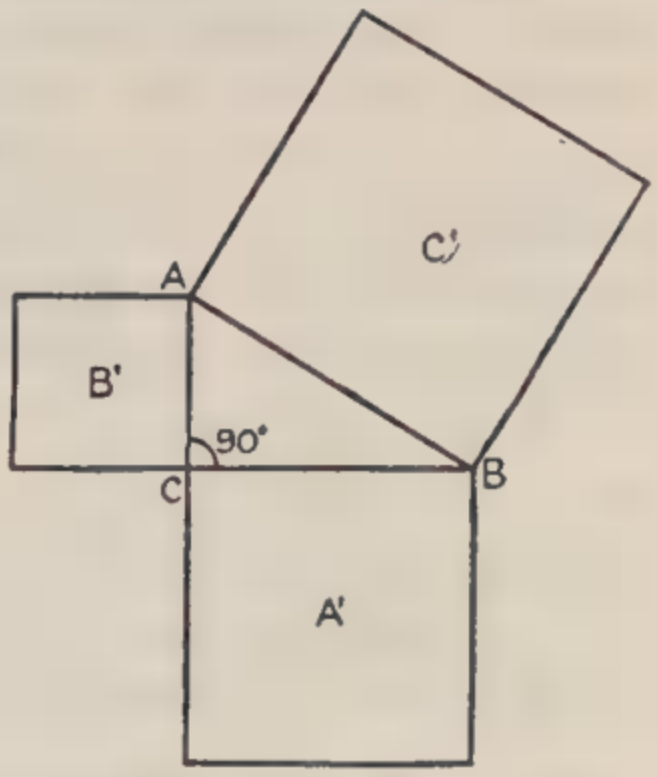

FIG. 1

The above rule is also the main metrical rule for co-ordinate geometry, which gives us the length of the line segment joining any two points. Consider, for example, two points in two dimensions, P_1 and P_2, whose

co-ordinates referred to a pair of axes in the plane are (x_1, y_1) and (x_2, y_2). By drawing the lines P_1Q and P_2Q parallel to the X and Y axes respectively, a right triangle P_1QP_2 is formed whose legs P_1Q and P_2Q are equal to x_2-x_1 and y_2-y_1 respectively, whence P_1P_2, the hypotenuse of the right triangle, or the distance s between the points, is equal to $\sqrt{P_1Q^2+P_2Q^2}$, or $s=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$. If we pass to indefinitely small quantities and choose to deal with differentials we have $ds^2=dx^2+dy^2$ where $dx=x_2-x_1$ and $dy=y_2-y_1$. Usually the physicists treat their differentials as very small quantities and we may do likewise, although this is not precisely what a differential represents.

In three dimensions similar formulae appear; namely, $s^2=x^2+y^2+z^2$ for the distance of a point from the origin and $s=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$ for the distance between two points P_1 (x_1, y_1, z_1) and P_2 (x_2, y_2, z_2), and also $ds^2=dx^2+dy^2+dz^2$, for the indefinitesimally small distance between two points.

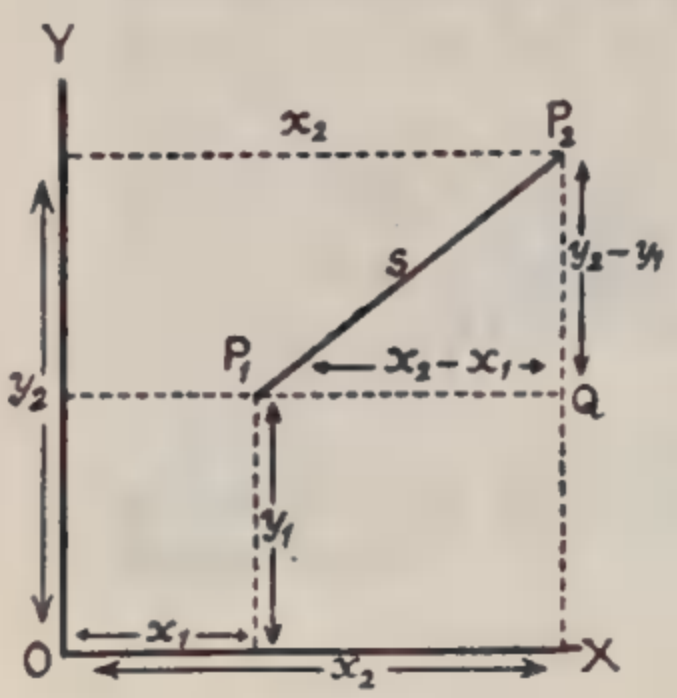

FIG. 2

In referring our geometrical entities to co-ordinate axes, or frames of reference, as they are called, we are interested in the properties of our geometrical entities and not in the accidental characteristics of our frames of reference, or the accidental characteristics of the form of representation we are using. Mathematicians discovered long ago that the form of representation is not of indifference to the results they obtain. Speaking roughly they have discovered that in one form of representation, they obtained characteristics $a, b, c, d, \ldots m, n;$ in another form, characteristics $a, b, c, d, \ldots p, q;$ and in still another form, characteristics $a, b, c, d, \ldots s, t$, . In cases where direct inspection was possible they find by checking up predicted characteristics, that such characteristics as a, b, c, d in our example actually belong to the subject of our analysis, whereas the characteristics $m, n, \ldots p, q, \ldots s, t, \ldots$, do *not* belong to our subject at all, but *vary* from one form to another depending on the form of representation. Such facts make mathematicians distinguish between characteristics which are *intrinsic*, which actually belong to the subject independently of the form of representation; and those which are *extrinsic*, which do not belong to the subject, but are accidental and vary with the form of representation we happen to use.

If we mix intrinsic and extrinsic characteristics we have a structurally distorted knowledge of our subject. Obviously we are interested in methods by which these two types of characteristics can be separated and distinguished.

Such methods are found in what we call the transformation of co-ordinates, which means the passing from one form of representation to another, from one system of co-ordinates to another which corresponds to translation from one language to another. Obviously those characteristics which are intrinsic to our subject are and must be *independent* of the accidental selection of our form of

representation, and therefore should remain unchanged when we pass from one frame of reference to another. Any characteristic which is changed by such a transformation of our systems of reference is clearly an extrinsic characteristic injected by the form of representation and not belonging to our subject; and so the transformation of co-ordinates is precisely the test we need and use.

Let us take for instance the line segment P_1P_2, as in Fig. 3. We may refer P_1P_2 to a system O, or to a system O'. Obviously the length of the line P_1P_2 is independent of the axes of reference used, and the formula for the length of a line is not altered, although the values for the x's and y's are different in the two systems. In other words, the sum of the squares of the differences of the co-ordinates remains invariant. In symbols,

$$s=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}=\sqrt{(x_2'-x_1')^2+(y_2'-y_1')^2}.$$

Such expressions, however, as x_1+x_2 or y_1y_2 . ; are *not* characteristics of our subject but characteristics of the particular frame of reference used, and so are mostly of no interest to us.

In such an elementary example as given here we are directly acquainted with our entities, and so we can inspect them directly and check for intrinsic and extrinsic characteristics. But when we deal with geometries of more than three dimensions, such checking becomes very difficult, if at all possible, and so new methods have to be invented.

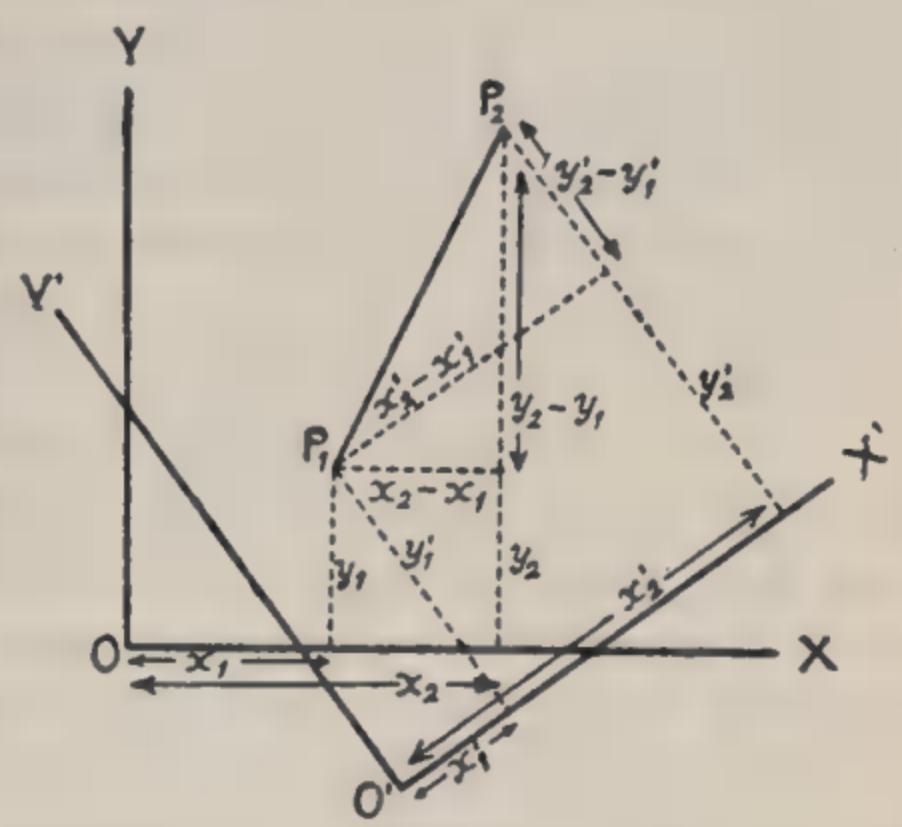

FIG. 3

If we wish to eliminate the unit by which we measure our lines, this can be done by using a relation called a *ratio*. Let us, for instance, select 3 points A, B, C, and write the invariant formulae for the distance AB and AC in the form $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$ and $\sqrt{(x_3-x_1)^2+(y_3-y_1)^2}$

then the *ratio* $$R=\frac{\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}}{\sqrt{(x_3-x_1)^2+(y_3-y_1)^2}}$$

is independent of our unit of measurement. If, for instance, this ratio $R=1$, we conclude that $AB=AC$, a characteristic which belongs to our lines and which is *independent not only of our system of reference but also of the unit which we have used.*

A great step forward in the formulation of methods which lead to invariant and intrinsic formulations was made in the invention of what is called the vector calculus and its extension in the modern tensor calculus. A few explanations of this principle will be of interest.

A vector is roughly a directed segment of a straight line on which we distinguish the initial and the terminal points. A vector has thus magnitude and direction. In practice we deal with two kinds of entities; some are purely

numerical, establishing a specific, mostly asymmetrical, relation and have no direction, as for instance, mass, density, temperature, energy, electrical charge, population, mortality , . These quantities which do not involve direction are called scalar quantities.

Such quantities as velocity, acceleration, electric current, stresses, flow of heat or fluids . , which involve not only magnitude but also a definite direction, are called vector quantities, and have given rise to a special calculus called the vector calculus.

The invention of the vector calculus was a most revolutionary and beneficial structural and methodological step. It was originated independently by Hamilton and Grassmann. The benefits of this method are manifold, but we are interested mainly in but two of them. The first is that vector equations are simpler and fewer in number than co-ordinate equations. The second, and most important, is that the language of vectors is independent of choice of axes, and of frames of reference. It is naturally invariant for any transformations of axes. If axes are needed we can easily and simply introduce them, but we always have means to discriminate between intrinsic and extrinsic characteristics. The modern tensor calculus which made the general theory of Einstein possible is simply an extension of the vector calculus.

The above methodological and structural remarks are of fundamental semantic importance to us in all our affairs. Human life and affairs are never free from linguistic issues. Their role is similar to that of *mathematics*, that is to say, a form of *representation* gives us not only the characteristics which are intrinsic in our subject, but also introduces extrinsic characteristics which do not belong to the subject of our analysis but are due to the particular language we use and its *structure*. The analysis of these linguistic issues is much belated and extremely difficult because of the structural complexity of our language. These issues were discovered first in mathematics because of its clear-cut structural *simplicity*; and it is important that we should be aware of such new and unexpected fundamental semantic problems. We will not enlarge upon this phase of the problem here, except to mention that the whole of the present work, which uses a different language, of a different *structure*, already shows the usefulness of the new method. Sometimes we discover new characteristics, and sometimes we are led to emphasize characteristics which are known but have not yet been sufficiently analysed.

To carry our linguistic analogy further, we may take, for instance, the statement, 'knowledge is useful'. We could translate this statement into any other language and it would preserve its meaning. But if we make the statement, 'knowledge is a word which has six consonants and three vowels', such a statement may be false when translated into another language. Mathematics, being a language, has difficulties similar to ordinary language, but in mathematics it is often much more difficult to separate from other statements those which are purely about the language used. The so-called tensor calculus attempts to perform this last task.

The tensor calculus is an extension of the vector calculus, which has become famous since Einstein. It gives us formulations independent of any special frame of reference. In using it we are automatically prevented from ascribing to the events around us characteristics which do not belong to them. The tensor equations give us absolute formulations, absolute being understood as relative, no matter to what. Obviously the only language fit to express the 'laws of nature' should be independent of the particular point of view or language of some observer. It should give us formulations invariant for any and all systems of reference, although we might use preferred systems of reference, as, for instance, the principal axes of an ellipse, without any danger. The reader should not miss the point that such an ideal should be considered as the highest ideal in science. It is the mathematical species of a theory of 'universal agreement'. The above *sounds* simple and innocent; but, when actually applied, plays havoc with most of our old 'universal laws'. These laws do not survive this important and uniquely valid test, and so become mere local gossip instead of being the 'universal laws' that they claim to be. We will return to the structural problem of invariant formulations later. At present we must explain some other simple considerations.

On any surface we need two numbers or 'co-ordinates' to specify the position of a point, and so a surface is called a two-dimensional manifold. Points in three-dimensional manifolds require three numbers; points in four-dimensional manifolds four numbers; and similarly for any number of dimensions.

For our purpose, it is enough to speak in two dimensions, as our statements can easily be generalized to any number of dimensions. If we want to localize a point on a surface it is enough to divide the surface into *meshes* by any two line-systems which cross each other. By labeling the lines of each system with consecutive numbers, two numbers, one from each system, will specify a particular mesh. If the meshes are small enough we will be able to locate any point accurately.

These specifying labels or numbers require that we know what kind of mesh we are using. *Distances* between points are *independent* of mesh systems.

For the above reasons it is important to have more data about the mesh system we are using, which means that we have formulae which express the *distance* between two points, which is independent of the mesh systems, in terms of the mesh system.

We have already seen, in our study of the differential calculus, that, as a rule, it is simpler to deal with very short distances, and that it is easy to pass to larger distances by the process of integration. As yet we have used only plane rectangular systems of meshes in our illustrations, but this restriction is not necessary. If we use oblique co-ordinates (Fig. 4), the formula for the elemental distance is $ds^2 = dx_1^2 - 2k dx_1 dx_2 + dx_2^2$, where k = cosine of the angle between the lines of partition.

The polar co-ordinates (Fig. 5) of the point P are the distance $r = OP$, of the point from the origin O, and the angle, $\theta = \angle XOP$, between the line OP

and the axis OX. The formula for the elemental distance in polar co-ordinates is $ds^2 = dr^2 + r^2 d\theta^2$.

Fig. 6 shows the co-ordinates frequently used by geographers; namely, geographic longitude and latitude, where the distance $ds^2 = d\beta^2 + \cos^2\beta d\lambda^2$.

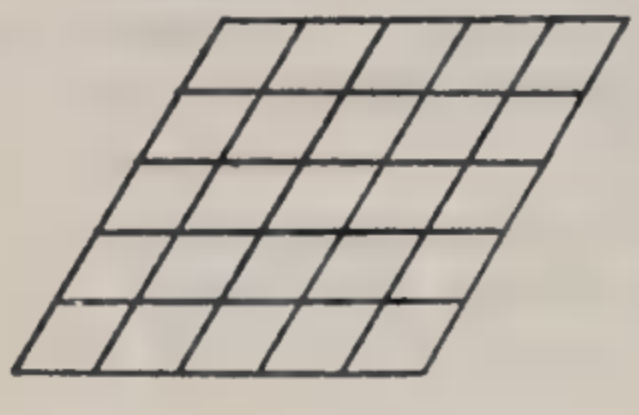

FIG. 4

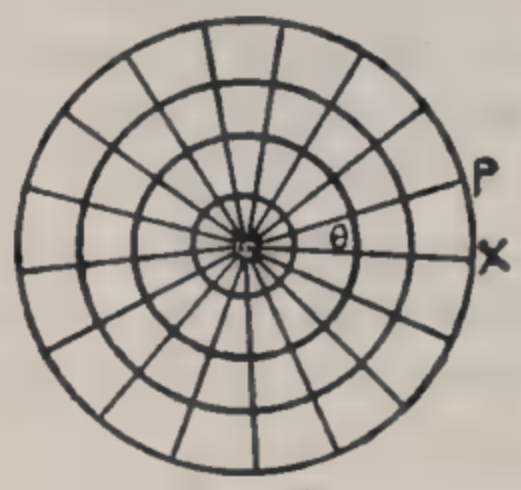

FIG. 5

FIG. 6

It should be noticed that these formulae for different systems of co-ordinates are *different*. To make it still more obvious to ocular inspection, we will tabulate them in one lettering, thus:

$$\begin{aligned} ds^2 &= dx_1^2 + dx_2^2 && \text{for rectangular systems} \\ ds^2 &= dx_1^2 + x_1^2 dx_2^2 && \text{for polar systems} \\ ds^2 &= dx_1^2 - 2k dx_1 dx_2 + dx_2^2 && \text{for oblique systems} \\ ds^2 &= dx_1^2 + \cos^2 x_1 dx_2^2 && \text{for latitude and longitude systems.}^1 \end{aligned} \tag{1}$$

It must be noticed that the values for the variables are not equal in these different equations. It is not necessary for the reader to know in detail how these formulae are obtained, but it is necessary to see that they are *different*, that they have different structure. The numbers of different co-ordinate systems we can use are infinite, but in practice we use only a few well-known types. There are also definite and simple formulae for passing from one system of co-ordinates to another.

We should not assume that in practice we always know what system of co-ordinates we are employing. For instance, before we learned that our earth is 'round', we did not know whether in our measurements we were employing the flat co-ordinates of a plane or spherical co-ordinates. We made some measurements and then we had to discover what kind of formulae would fit these measurements.

To find out what kind of co-ordinate system we are using, we select two points, let us say (x_1, x_2) and (x_1+dx, x_2+dx) very close together, make our measurements of ds, and then test our ds to find which formula it fits. If we find for instance that our ds^2 is always equal to $dx_1^2 + dx_2^2$ we may assume for simplicity and our purpose that our co-ordinate system is plane and rectangular.

If our measurements fit any of the first three formulae (1), we may assume for simplicity and our purpose, that we are dealing with a plane surface, as each of these systems belongs to the plane. But if we find that the actual measurements of ds^2 are such that they never fit these first three formulae, but only the fourth one, we know, that our surface is not plane but curved

like a sphere. Try as we may, we shall be unable to build on a plane any co-ordinate system which will fit the last formula. Thus we arrive at an important conclusion; namely, that from *measurements* we have a *structural* hint as to the *kind of world* we are in.

Section B. On the notion of the 'Internal Theory of Surfaces'.

Let us imagine some two-dimensional beings confined to their surface and unable to have a look at that surface from our third dimension. For them our third dimension would be 'unthinkable', and therefore the surface of a sphere like our earth which is curved in the third dimension would also be 'unthinkable' or 'beyond them'. Should they conduct some measurements in their 'world' and find that these measurements did not fit any of the first three formulae but only the fourth, they would have to reconstruct radically their 'world conception' and conclude that their world was a spherical surface. Our own situation does not differ radically from the situation of the inhabitants of this hypothetical two-dimensional world.

If we find ourselves so restricted as not to know whether we are finally dealing with a flat or spherical surface, we can select a point O and with a definite radius R describe from this point a circle ABC. Then we can measure the circumference of this circle. Now we know from geometry that in *the plane* the circumference of the circle $L=2\pi R$ where R is the radius of the circle and $\pi=3.1415\ldots$ If our surface is flat ($ABCD$), our measurement of L and R will satisfy the relation expressed in the formula. But if the surface is curved, our $R=OA$ will be larger than $R'=AO'$, and we shall find that our π is not 3.1415 . . . , but smaller. We see once more that the *metrical properties of our world throw some light on its structural character.*

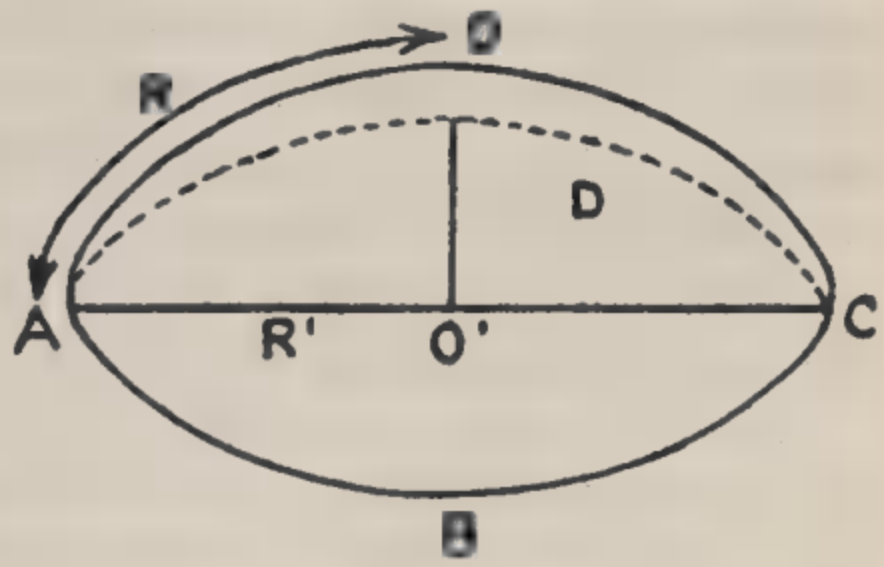

FIG. 7

We should notice also that the curvature of a two-dimensional surface is in the third dimension and that it is the means of giving us data about the surface without our leaving the surface and going into a third dimension. It is easy to convince oneself about these facts by taking 12 wires or strings of equal length and constructing the figure shown in Fig. 8. If we build it on a flat surface the 12 equal wires will fit exactly. But if we try this experiment on a curved surface, for instance on a pillow-, or saddle-shaped surface, the last closing wire will not fit, and will be too short or too long depending on the kind of surfaces we have.

FIG. 8

The formulae (1) have been generalized to

$$ds^2 = g_{11}dx_1^2 + 2g_{12}dx_1dx_2 + g_{22}dx_2^2 \tag{2}$$

for two dimensions and to

$$\begin{aligned} ds^2 = {} & g_{11}dx_1^2 + g_{22}dx_2^2 + g_{33}dx_3^2 + g_{44}dx_4^2 + 2g_{12}dx_1dx_2 \\ & + 2g_{13}dx_1dx_3 + 2g_{14}dx_1dx_4 + 2g_{23}dx_2dx_3 + 2g_{24}dx_2dx_4 + 2g_{34}dx_3dx_4 \end{aligned} \tag{3}$$

for four dimensions. It is easy to see that $ds^2 = dx_1^2 + dx_2^2$ is obtained from (2) by taking $g_{11} = 1$; $g_{12} = 0$ and $g_{22} = 1$. This applies to formula (3) out of which we can have any of the other formulae by equating some of the g's to zeros, or to one or to other values. Formula (3) is called 'the generalized pythagorean rule', of which the ordinary form as given previously is only a particular case. We see, by comparing the formulae (1) with (2) and (3), that these g's are not equal for different systems of co-ordinates, and that they are factors in measure-determination which represent the geometry of the surface considered. It is customary to write the above formulae in an abbreviated form: thus, $ds^2 = \sum\sum g_{mn}dx_mdx_n$, where we give to m and n the values 1, 2, 3, 4, or, $(m, n = 1, 2, 3, 4)$ and where the symbol $\sum$ means summation.

We will now explain briefly the above generalizations and the meaning of the g's given in the expressions.

In the beginning of the nineteenth century the mathematician Gauss formulated the *internal* theory of surfaces without reference to the plenum in which they are embedded. This theory perhaps is and will remain a model on which all theories should be built. He introduced also a new kind of co-ordinates which have become of paramount importance, and which since Einstein are called gaussian co-ordinates. Gauss investigated the theory of surfaces, which are in general curved, embedded in three-dimensional 'space'. In 1854 the great mathematician Riemann generalized the two-dimensional gaussian theory to a continuous manifold of any number of dimensions. Historically, both Gauss and Riemann can be considered as the precursors of Einstein.

Let us imagine a surveyor to have the task of mapping a thickly wooded hilly region. Because of the conditions of his work, he can not use optical instruments, and he has no 'straight lines' to deal with. So euclidean geometry will, in general, not be applicable to the region as a whole. It can be assumed, however, that euclidean geometry may be applied to *very small* regions which can be considered flat. What we know already about the differential and integral calculus shows us that such approximations, when taken on a very small scale, are perfectly reliable and justifiable.

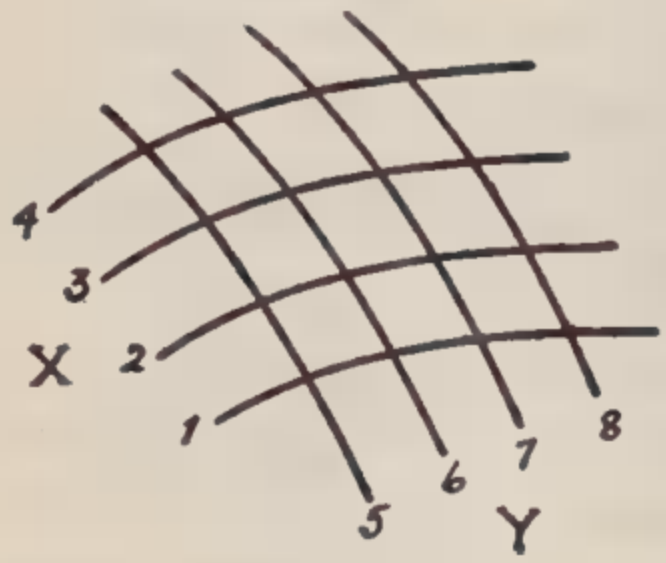

FIG. 9

The surveyor would lay out on his ground a network of smoothly curving lines, in two families, an X family and a Y family. (Fig. 9.) All the curves of the X family would intersect all the curves of the Y family but no X curve would intersect another X curve, nor a Y curve another Y curve.

Let us take the surveyor's network and label the curves by consecutive numbers in each family. The essential point is that these numbers, (let us call them the X and Y numbers) do *not* represent either lengths, or angles or other measurable quantities, but are simply labels for the curves, much as when we label streets by numbers.

But such numbering does not lead us far. We must introduce some *measure relations*. We have at our disposal a measuring chain and the *arbitrary* meshes of the network which we have introduced. The next step is to measure the small meshes one after another and plot them on our map. When this is done we have a complete map similar in structure to our region. Because of the smallness of the meshes we can consider them as small parallelograms, and such parallelograms can be defined by the lengths of two adjacent sides and one angle.

We may, however, proceed differently, as shown on Fig. 10.

Let us select one mesh, for instance the one bounded by the curves, 3 and 4, and by the curves 7 and 8. Let us consider a point P within the mesh, and let us denote its distance from the point O $(x=3,\ y=7)$ by s. This distance could be directly measured. Let us draw from the point P parallels to our mesh lines and label the intersections with mesh lines by A and B, respectively. Let us also draw PC perpendicular to the x axis.

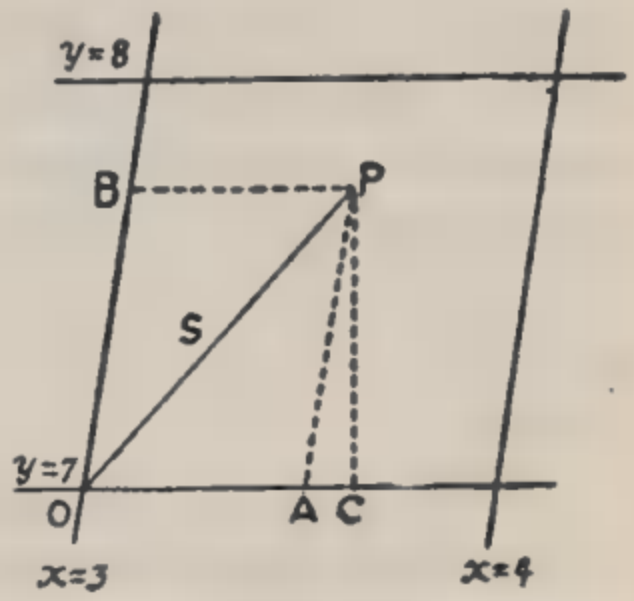

FIG. 10

The points A and B then also have numbers or labels or gaussian co-ordinates in our network. The co-ordinate of A may be determined by measuring the side of the parallelogram on which A lies and the distance of A from O. We can regard the relation called the *ratio* of these two lengths as the *increase* of the x co-ordinate of A towards O. We shall denote this increase itself by x, choosing O as the origin of the gaussian co-ordinates. Similarly, we determine the gaussian co-ordinate of y of B as the ratio in which B cuts the corresponding side. We see that these two *ratios*, which for brevity we call x and y, are the co-ordinates of our point P.

As x and y are *ratios* they of course do not give us the lengths of OA and OB but the lengths are given, for example, by ax, and by, where a and b are definite numbers, to be found by further measurements. If we move the point P about, its gaussian co-ordinates change but the *numbers* a and b which give the ratio of the gaussian co-ordinates to the true lengths *remain unchanged.*

We find the length, s, which is the distance of the point P from O, from the right triangle OPC by the pythagorean rule: $s^2=OP^2=OC^2+CP^2$. But $OC=OA+AC$ and therefore by substituting and squaring we have $s^2=AO^2+2OA\cdot AC+AC^2+CP^2$. From the right triangle APC we have $AC^2+CP^2=AP^2$, whence substituting again we have $s^2=OA^2+2OA\cdot AC+AP^2$. But $OA=ax$, $AP=OB=by$, and as AC is the projection of $AP=by$, it also has

a *fixed ratio* to it, whence we may put $AC=cy$, and so we obtain the important formula $s^2=a^2x^2+2acxy+b^2y^2$, in which a, b, c are *ratios* given by *fixed numbers*. Usually this formula is represented differently, a^2 is designated by g_{11}, ac by g_{12}, and b^2 by g_{22}; whence our formula becomes $s^2=g_{11}x^2+2g_{12}xy+g_{22}y^2$ in which the numbers 11, 12 and 22 are simply ordering labels without quantitative values, mere subscripts, labels, indices . , which indicate that the different g's have different values. We see that the above formula is the one which was given previously by (2).

The g's with different labels serve just as sides or angles for the determination of the actual sizes of the parallelograms and we call them the *factors* of the measure *determination*. They may have different values from mesh to mesh, but if they are known for every mesh, then, by the last formula, the true distance of an arbitrary point P, within an arbitrary mesh from the origin can be calculated.[2]

The procedure by which we can locate any point on the surface is simple. If our point P is between the two curves $x=3$ and $x=4$ we can draw nine curves between these two curves and label them 3,1; 3,2; . . . ; 3,9. If P now lies between curves 3,1 and 3,2 we can draw nine curves between these two curves and label them 3,11; 3,12; . . . ; 3,19 , . We could do similarly with the y curves and in this way we would succeed in assigning to any point as accurate a pair of numerical labels as we pleased, and so finally have the gaussian co-ordinates of any point. We used nine curves simply to get the very convenient decimal method of labeling. The cartesian co-ordinate systems which we use in plane geometry obviously represent only *special cases* of gaussian systems.

As we have already seen, our g's are *ratios*, and so represent numbers. Such numbers may be regarded as tensors of zero rank for convenience of the mathematical treatment; and the quantities g_{xx}, g_{xy}, g_{yy}, may be treated as components of a tensor. Since this tensor determines the measure relations in any particular region, it is called the *metric fundamental tensor*. Its value must be given for the region in which we want to make our calculations. It determines the full geometry of the surface in a given region; and, conversely, we can also determine the fundamental tensor in a given region from measurements made in that region, without any previous knowledge of how our curved surface is embedded in 'space' at the place in question. The fundamental tensor in general varies continuously from place to place, and so every geometric manifold may be regarded as the field of its metric fundamental tensor.

Purely mathematical investigations show that the fundamental tensor defines a number called the 'Riemann scalar', which is completely independent of the co-ordinate system and leads to the definition of the *curvature tensor*, which can be connected with the 'matter tensor'.[3]

The main importance of the introduction of such arbitrary curves is to produce formulae for the surfaces which remain unaltered for a change of the gaussian co-ordinates—in other words, which remain invariant. This was achieved by the introduction of the relations called *ratios* which are pure

numbers, and so the geometry of surfaces becomes a theory of invariants of a very general type.

On curved surfaces there are in general no straight lines—there are *shortest lines*, which are called 'geodetic lines'. To find them, we divide any arbitrary lines joining two points into small elements, which we measure, and select the line for which the sum of these elements is less than for any other line between the two points.* Analytically we can calculate them, when the g's are given, by the aid of the generalized pythagorean theorem. The geodetic lines, and also the curvature, are given by invariant formulae, which represent intrinsic characteristics of the surface, independent of any co-ordinates. All higher invariants are obtained from these invariants.[4]

We shall not attempt to give an explanation of the tensor calculus, as at present there is no elementary means of presenting a brief explanation; short of a small volume—at least the writer does not know of any.[5]

The name 'tensor' originally came from the Latin word *tendere* = to stretch, whence *tensio* = tension. Nowadays, however, it is used in a more general way; namely, to express the *relation* of one vector to another, and not necessarily to imply stress or tension. As an example, we can give the representation for stresses occurring in elastic bodies, which originally led to the name.[6]

As we have already seen, when we deal with *relations* of vectors our expressions become additionally independent of units. Such equations, independent of the measure code, are called tensor equations.[7]

As we are interested in equations which are invariant under arbitrary transformations, certain functions, called tensors, are defined, with respect to any system of co-ordinates by a number of functions of these co-ordinates, called the components of the tensor, from which we can calculate them for any new system of co-ordinates. If two tensors of one kind are equal in one system, they will be equal in any other system. If the components vanish in one system, they vanish in all systems. Such equations express conditions which are independent of the choice of co-ordinates. By the study of structural laws of the formation of tensors we acquire means of formulating structural laws of nature in generally invariant forms. Obviously, such methods and language are uniquely appropriate for physics and the formulations of the laws of nature. If a law cannot be formulated in some such form, there must be something wrong with the formulation and it needs revision.

The tensor calculus is also peculiarly fitted to describe processes in a *plenum*. We do not use it to describe the metrical conditions but to describe the *field* which expresses the physical states in a metrical plenum.

Eddington gives an excellent example of the fact that it is definitely necessary to look into the way we build up our formulae (structure) and the method of handling them.

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*More generally, the geodetic represents a track of minimum or maximum interval-length between two distant events, either of them being unique (one-valued) in a given case.

The problem is to determine whether a particular kind of space-time is possible. We must investigate the different g's which give us different kinds of space-time, and not those which distinguish different kinds of mesh systems in one space-time. This means that our formulae must not be altered in any way if we change the mesh system.

The above condition makes an extraordinarily simple test of laws that have been or may be suggested. Among others, Newton's law is swept away. How this happens can be shown in two dimensions.

If in one mesh system (x,y) we have $ds^2 = g_{11}dx^2 + 2g_{12}dxdy + g_{22}dy^2$, and in another system (x',y') $ds^2 = g_{11}'dx'^2 + 2g_{12}'dx'dy' + g_{22}'dy'^2$, one law must be satisfied if the unaccented letters are replaced by accented letters. Let us suppose that the law $g_{11} = g_{22}$ is assumed. We change the mesh system, for instance, by spacing the y lines twice as far apart, that is, we take $y' = y/2$ and keep $x' = x$. Then $ds^2 = g_{11}dx^2 + 2g_{12}dxdy + g_{22}dy^2 = g_{11}dx'^2 + 4g_{12}dx'dy' + 4g_{22}dy'^2$. We see that $g_{11}' = g_{11}$ and $g_{22}' = 4g_{22}$. Whence if g_{11} is taken equal to g_{22}, g_{11}' *cannot* be equal to g_{22}'.

A few examples would convince us that it is extremely easy to change a formula entirely by the mere change of mesh systems. It seems unnecessary to emphasize the fact that 'universal laws', to be 'universal', should not depend structurally to such an extent on the accidental and, after all, unimportant, choice of reference systems.[8]

To remedy such a state of affairs, impossible in mature science, the tensor calculus was invented. The whole general theory of Einstein seems to demand that the equations of physics should ultimately be expressed in tensor forms; in other words, that 'universal laws' should cease to be 'local gossip'; a demand which must be granted, and *on this point* the Einstein theory is beyond criticism and is an epochal methodological advance of an irreversible structural linguistic character.

Section C. Space-time.

FIG.–11.

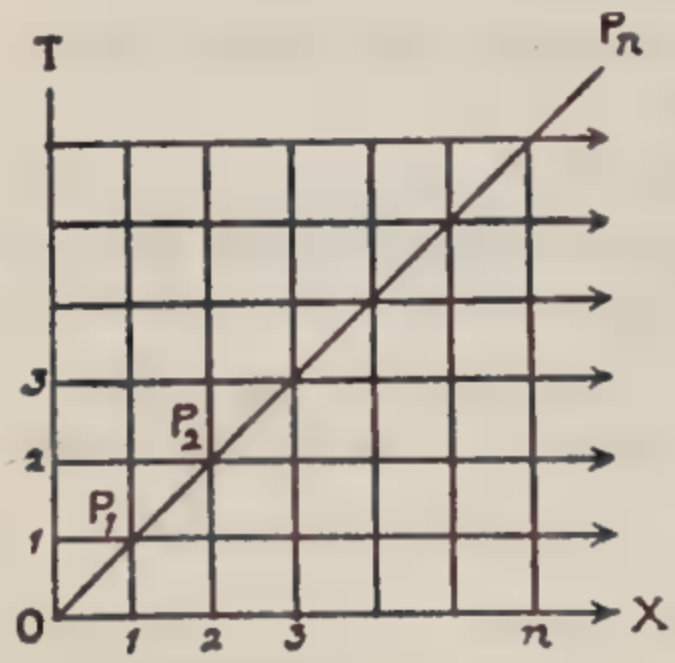

FIG. 12

In dealing with co-ordinate systems we have heretofore used them to represent only 'spatial' entities, spreads of different dimensions. It is desirable to become acquainted with a different use of co-ordinates, in which one of them will represent 'time'. The last use is just as simple as the former, but the graphs which we obtain are different.

Let us take the simplest example, of a point P moving uniformly along a straight line OX with the velocity of one inch per second. We could represent its movement in one dimension, as in Fig. 11, and *say* that our point P is at P_1 after one second $(t=1)$, at P_2 after two seconds $(t=2)$. ; at the point P_n after n seconds $(t=n)$.

But we could also represent this movement in a different way. We could choose two mutually perpendicular axes OX and OT as in Fig. 12, OX representing the 'spatial' actual direction of the movement and OT, which we have heretofore used to represent a second 'spatial' co-ordinate, now representing the 'time' co-ordinate.

We would lay off on the X axis our inches, 1, 2, 3, . . . n, and on the T axis our seconds 1, 2, 3, . . . n. In our two-dimensional *space-time* our point P would be at the point O ($x=0$, $t=0$). After one second it would be at the point P_1 ($x=1$, $t=1$), after two seconds at the point P_2 ($x=2$, $t=2$) . ; after n seconds at the point P_n ($x=n$, $t=n$). We see that the position of our *point* P in two-dimensional *space-time* would be represented by a series of points each given by two data: one 'spatial', the other a corresponding 'time'. If the intervals are taken indefinitely small, in the limit our *'moving' point would be represented by a static line inclined to the X axis*. We could then speak either of our 'moving' point, or else *not use* the term 'moving' but speak of infinitely many *static points*, each given by two numbers, one representing a distance, the other 'time'. Our 'moving' *point* would become a *static world-line*. The reader should notice that in this case we have structurally changed our *language* from dynamic to static, and raised the dimension. Our mathematical 'moving' 'point', which had no dimension in our one-dimensional 'space', is in our two-dimensional *space-time* represented by a *static* one-dimensional *line*.

In this example we had uniform translation. We did not introduce acceleration. The distances were proportional to the 'times', hence our line was 'straight' and inclined to the X axis at a constant angle.

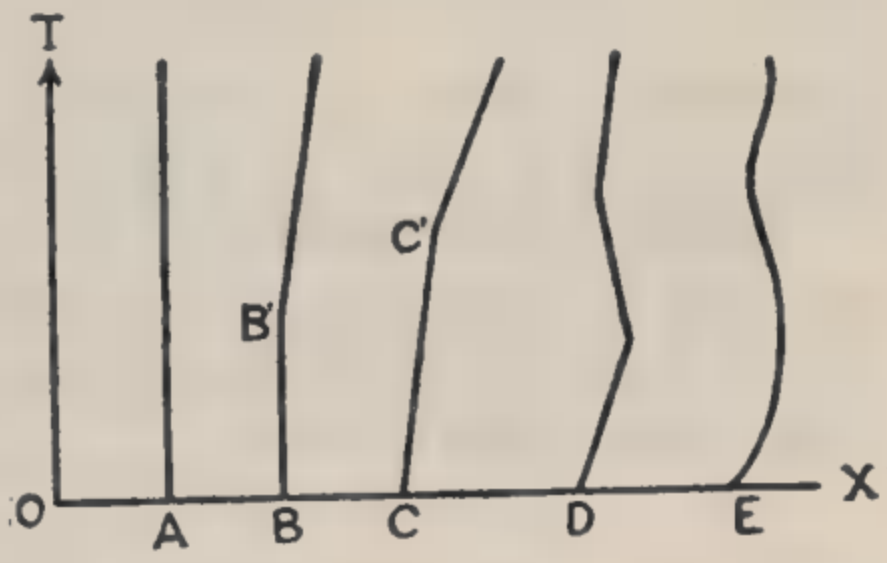

FIG. 13

Using such space-time representation we see that a point when it is not 'moving', but is stationary, is represented by a line parallel to the 'time' axis T, as shown at A on Fig. 13. Our point A is getting older, so to speak, but does not 'move'. In the next case, the point B does not 'move' until it is some seconds old, when at B' it begins to 'move' with constant velocity. Point C 'moves' in the beginning at one constant velocity until C' where it acquires a certain different velocity and the direction changes.

In Fig. 13, D represents a point experiencing a series of sudden changes of velocities. The graph is a succession of short straight lines forming a broken line or open polygon. As the changes of velocity occur more and more frequently the sides of our polygon become smaller and smaller; and in the limit, as the changes of velocity become continuous, our broken line becomes a smooth curve E.

Motion with continuously changing velocity is called accelerated or retarded motion. The rate of change of velocity is called acceleration and is

represented by the second derivative of the distance with respect to the 'time'; symbolically, $A = dv/dt = d^2s/dt^2$.

It is important to notice that in space-time an *accelerated* motion is represented by a *curved* line. In uniform (constant velocity) motion the distances are proportional to the 'times', and the line is straight and its equation is of the first degree. In accelerated motion the distances are *not* proportional to the 'times', the lines are curved and the 'time' element dt enters in the second degree at least; namely, as dt^2.

For example, let us study the graph of the motion represented by the equation $x = At^2/2$ which means that the distance x is proportional to the square of the 'time'.

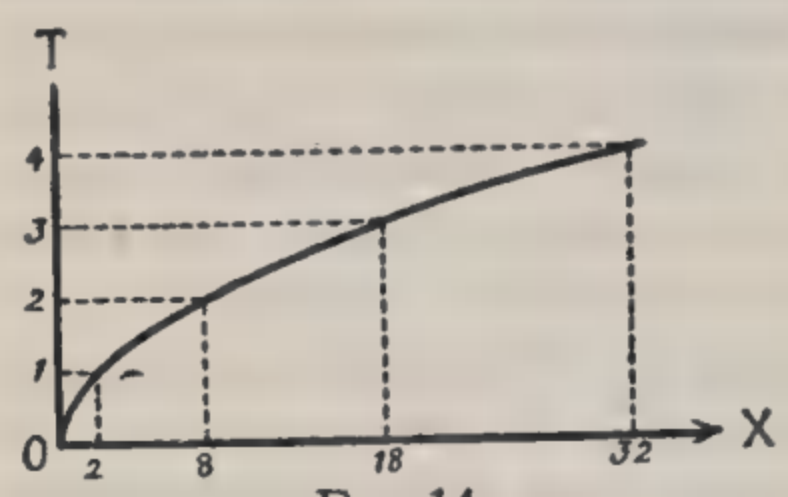

FIG. 14

Let OX (Fig. 14) be the 'spatial' axis, and OT the 'temporal' axis. We lay off on our T axis equally-spaced points, representing the seconds 1, 2, 3, 4 . , and calculate the distances x for each of these values from the equation $x = At^2/2$ where A represents a constant acceleration.

Let us assume that the constant acceleration A is given as 4 metres per second per second. The equation $x = 4t^2/2$ becomes $x = 2t^2$. Corresponding to the values $t = 0, 1, 2, 3, 4$. , we have the values $x = 0, 2, 8, 18, 32$, . If we plot these points, and assume that the change is continuous, we may join the points by a continuous *curve*, which represents the motion of the point as a *curved* world-line.

Similarly, in three-dimensional space-time, a point moving uniformly in the *plane* XY would be represented in the *plane* XY by the *line* AB, and in three-dimensional space-time by the static line AB', where the 'times' are proportional to the distance.

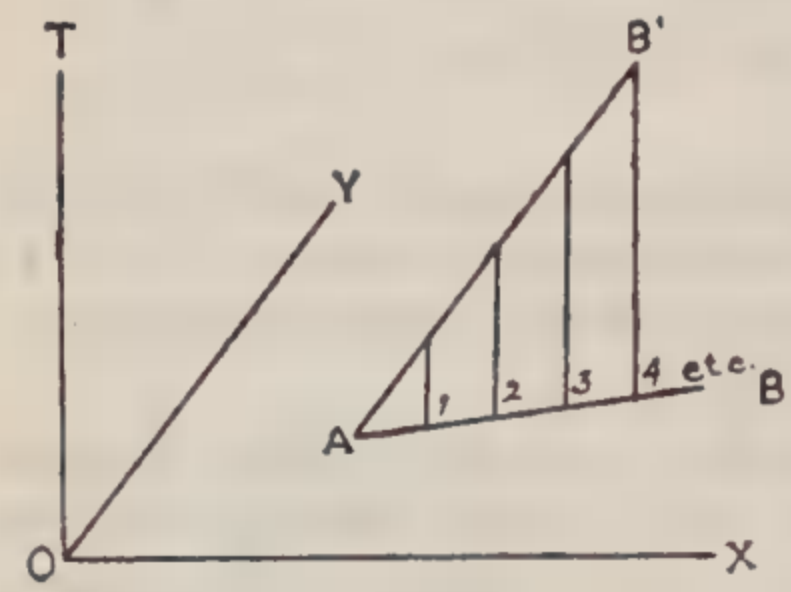

FIG. 15

As we have already seen, non-rectilinear motion may be considered as accelerated motion. We will generalize the above to the case where any *curved* path is traversed with *constant* velocity. In this case the direction of the velocity is changed. If we take the motion of a point which describes a circular orbit with *constant* velocity, it is easy to find its acceleration, which is called in this case centripetal acceleration.

Let us consider a point P, moving in a circular orbit with a *constant* velocity v, as given in Fig. 16. If at a certain 'time' it is at A, after a short interval t, it will be at B. The direction of the velocity will be changed from AA' to BB'.

If we construct the triangle DCE by drawing CD parallel and equal to AA', and CE parallel and equal to BB', we see that the angle $\angle DCE$ is equal to the angle $\angle A'A''B'$ because the sides are parallel, and it is also equal to the angle $\angle AOB$ whose sides are perpendicular to AA' and BB'. The triangles ABO and CDE are similar because they are isosceles and the angles between the equal sides are equal. Clearly the side $DE=w$, represents the supplementary velocity which transforms AA' into BB'. We know that in similar triangles the sides are proportional so we can write $DE/CD=AB/OA$. By inspection of our figures we see that $DE=w$; $CD=v$; $OA=r$, the radius of the circle. The chord AB may be taken as the arc AB of the circle, provided the 'time'-interval is taken sufficiently small. Let us write chord $AB=s$. We have $w/v=s/r$ or $w=sv/r$. If we divide both sides of our equation by t we have $w/t=sv/tr$. But $w/t=A$, the acceleration, and $s/t=v$ hence $A=v^2/r$. In words, the centripetal acceleration is equal to the square of the velocity in the circle divided by the radius.

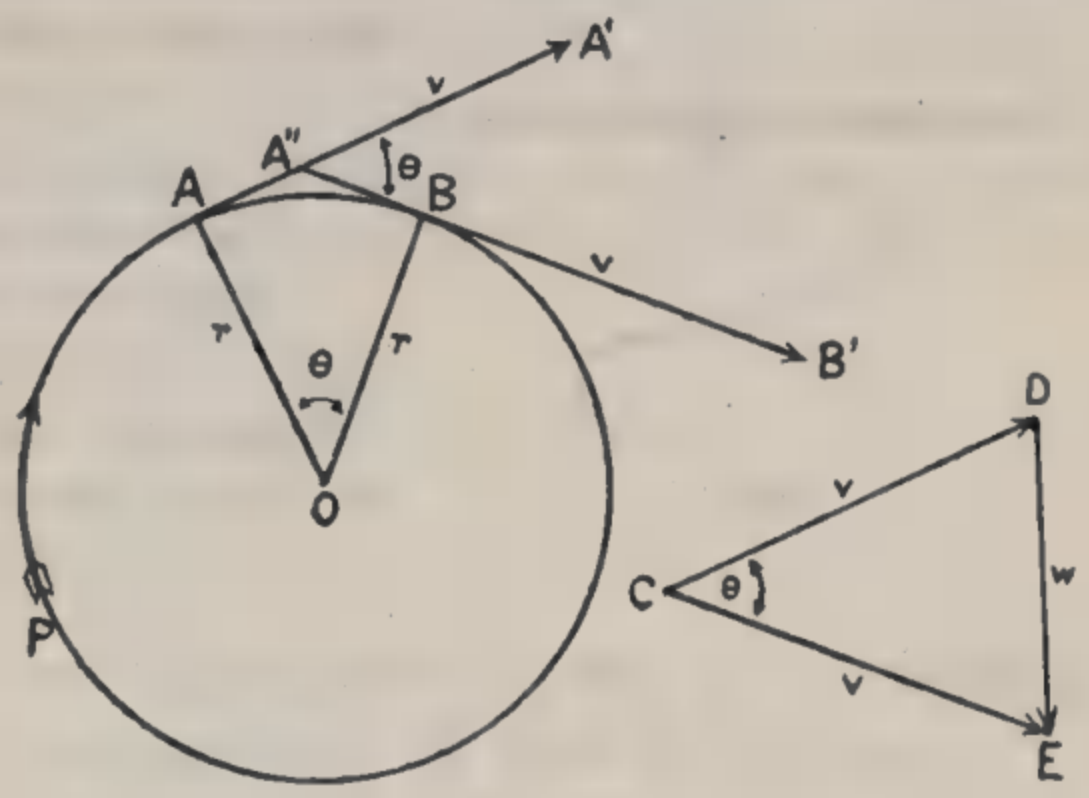

FIG. 16

The above formula is of structural importance because it is the foundation for the empirical proof of Newton's law of gravitation. For our purpose it is important for other reasons, to be stated later.

There are two more diagrams which should be considered, in this connection. Fig. 17 represents the plane circular motion of a point P whose orbit in the plane XY is the circle PAB. In three-dimensional space-time the plane circular orbit of motion would be represented by the static cylindrical helix (or screw-line) with axis parallel to the 'time' axis T. (Fig. 17.) We should note that the motion is *dynamically circular* in the XY plane, yet a three-dimensional space-time representation gives us a stationary helix.

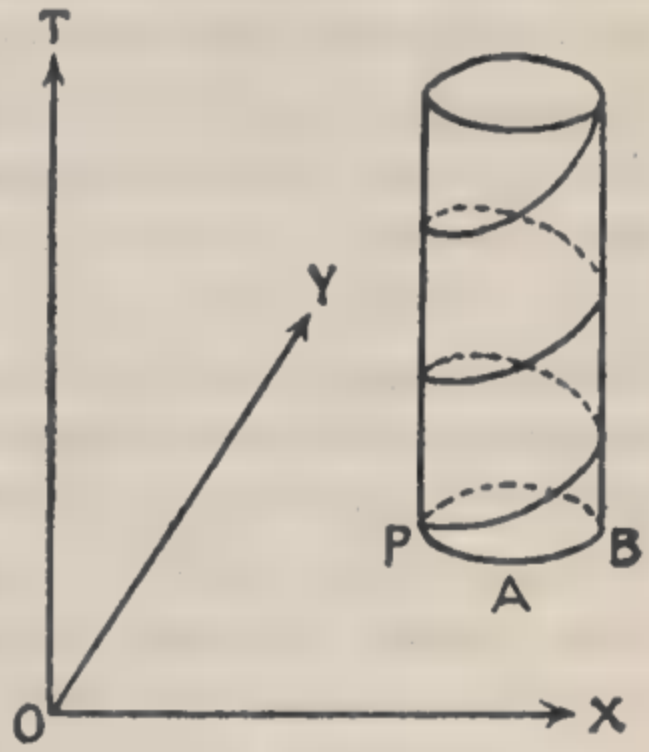

FIG. 17

Similarly for vibrational movements which could be represented in one dimension by to-and-fro movements on the X axis from A to B and from B to A. (Fig. 18.) If we introduce our space-time form of representation by introducing the T axis, our vibrational world-line would be represented structurally by a wave-line along the T axis. In particular, if the vibrational motion is simply harmonic, a proper choice of the 'time' unit makes the wave-line a sine curve.[9]

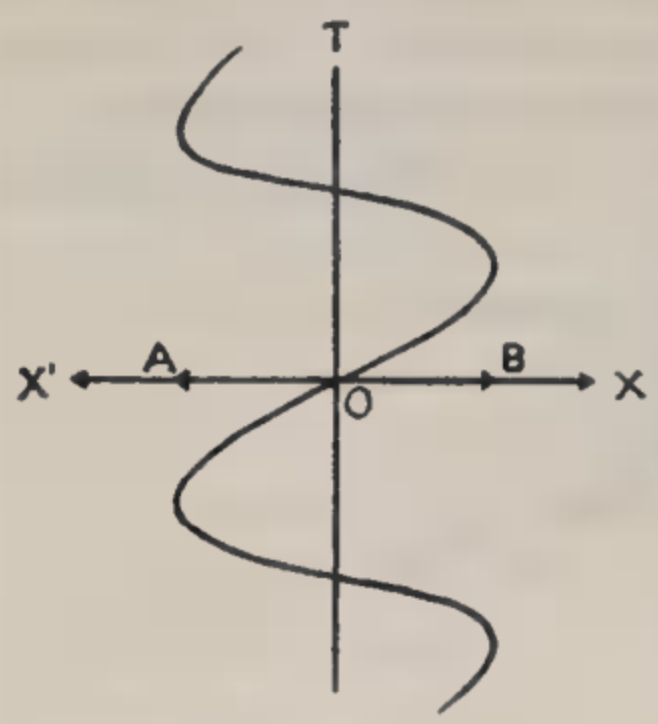

FIG. 18

Becoming thoroughly familiar with these few simple examples takes away a great deal of mystery from the Minkowski-Einstein and the new quantum world. We see that after all there is nothing extraordinary in the fact that in languages of different structures we get different forms of representations and pictures, and that in a world where accelerations abound we may very profitably use the term 'curved'.

When we come to speak about the Einstein theory, the four-dimensional space-time world of Minkowski, and the new quantum mechanics, we shall have considerable use for these few notions and illustrations.

Section D. The application of geometrical notions to cerebral localization.

In the present work we are dealing in the main with structure and the adjustment of the structure of our languages to empirical structures, and at this point it will be of use to suggest some of the consequences which follow from what has been said.

The question of cerebral localization is a difficult and vital problem. In former days it was supposed that the brain had individualized centres with strictly defined functions. Attempts were made to ascribe to definite cerebral parts definite functions such as 'memory', 'intelligence', 'morality', 'talents' , . In the meantime, experimental facts disproved such structural views, and as a reaction another tendency appeared; namely, to deny any localization.

Modern researches show unmistakably that both of these extreme tendencies are at variance with experimental structural facts. It appears that the lower centres play a more important role according as the terminal, or higher centres, are less developed and that there is considerable variability, at least in man, not only from the morphological and histological aspects but also from the functional aspect. It was found impossible to generalize from the particular development of centralization and functional distribution in one species to the distribution in another species. Localization may vary even in one individual under different circumstances.[10] Metabolism, and slight disturbances in the functioning of a neuron, were also found to have a most far-reaching influence, shown in its relations to other groups of neurons. The problems of localization are far too complex to attempt even an account of them, the more so since the reader will find excellent accounts of them in the large literature on the subject. The general conclusion reached by practically all investigators is that some localization of nervous function does exist, yet it has a certain variability which depends on an enormous number of factors.

The methods explained in this chapter will enable us to suggest a method by which we can orient ourselves in the bewildering complexity of the functioning of the nervous system.

One of the main difficulties is that the structure of this world is such, that it is made up of absolute individuals, each with unique relationship toward environment (in the broadest sense); and we have to speak about it in terms of generalities. 'Laws' . , formulated in the *old two-valued ways*, can never account adequately for the facts at hand, being only approximations. The mathematical methods which have already been explained give us at once a great advantage. We have seen that if we have a function, $y=f(x)$, let us say, and take the graph of this function, to every point of the graph there corresponds a pair of values x and y. We have seen also that each of the four quadrants I, II, III, IV has a characteristic pair of signs. In quadrant I; both x and y are positive; in II, x is negative and y positive; in III, both x and y are negative; and finally, in IV, x is positive and y negative. We can easily see that the value of the variables may be thought of as variable conditions different for each individual, and that definite *localizations* correspond to them. In our example we had to do with a function of one independent variable, and we had a one-dimensional line, curved in two dimensions. When we had a function of two independent variables we had a surface, which in general was curved in a third dimension. By analogy we may pass to any number of dimensions, where by dimension we do not mean anything mysterious, but roughly the number of variables involved in the problem.

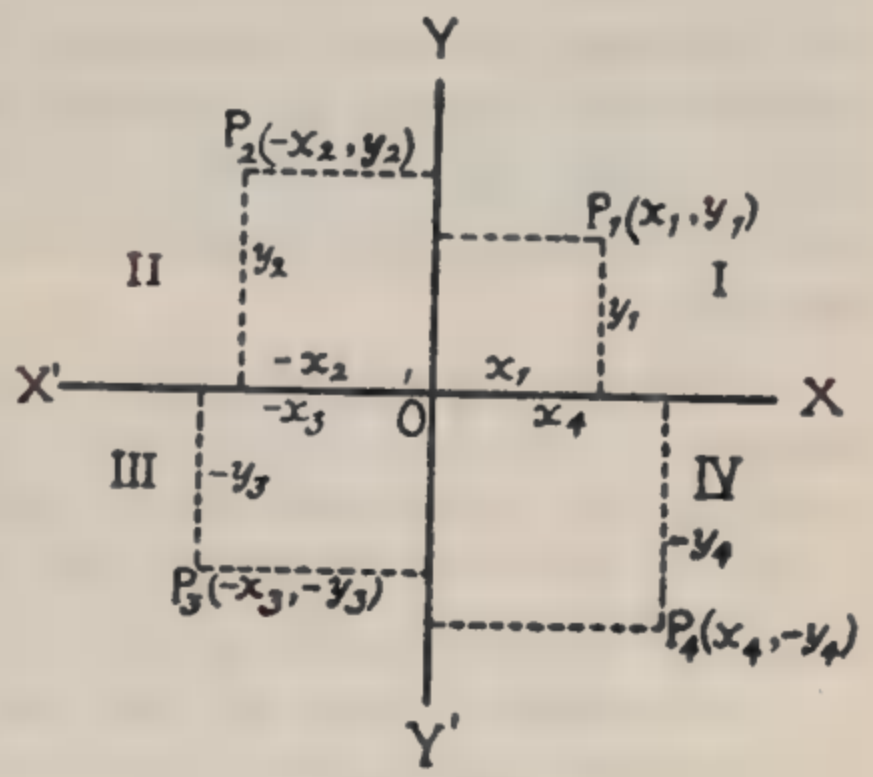

FIG. 19

We see that if we think of the activity of the nervous system in terms of a mathematical function with an enormous number of variables, we shall not only have place for the uniqueness of each individual, determined by the value of the variables and the character of the function, but that this would also imply a *localization*, which is *permanent* in a given individual at a given 'time'; which again implies the totality of 'circumstances' , . Our function would be $N=f(x_1, x_2, x_3, \ldots x_n)$.

In fact it is hard to see how it is possible to analyse the activities of the nervous system in any other way. The facts are, that every organism is an individual, distinct and different from others, and so we must have means to take this individuality into account. Different values for different variables take care of this point. Similarities are accounted for by the general structural character of the functions. For instance, any quadratic equation with two unknowns gives us a conic section. An equation of the type $y^2=ax$ represents a parabola, the graph of any equation of the form $xy=a$ represents a hyperbola , . For every definite set of values of our variables the implied localization is also definite, which corresponds to the fact that in a given individual at a given 'time' . , the localization is definite. One value for the whole function can be

reached by giving different sets of values to the different variables. For instance, in the function $z=5x-2y-1$, if $x=1$ and $y=1$, then $z=2$; but we can have $z=2$ by taking $x=2$ and $y=3.5$. Or if one of the conditions be non-existent, which means that the value of one of our variables is zero, for instance, $x=0$, we still could have the value $z=2$ by taking $y=-1.5$. This fact accounts for the *many-to-one* correspondences of causal factors, typified, for instance, by sunshine *or* cod liver oil producing a similar effect.

It should be understood that in what is said here, the numerical values do not matter. In most of the cases we are not advanced enough to be able to deal with such numerical values. What is to be emphasized is the structure of the language we use. The method should enable us, instead of dealing with generalizations in the old language, which somewhere have to be contradicted, to use a language of mathematical structure which shall account for the facts and leave room for the great individual varieties of organisms in structure and function.

After all, we should not be surprised that the theory of functions and language of functions is structurally appropriate in expressing, and so in understanding, the *functioning* of the nervous system, or any other system. Personally I have benefitted greatly through this method; and many baffling structural complexities have been much simplified.

Structurally, when we use the language of functions, variables . , we automatically introduce *extensional* structure, as already explained, and we have at our disposal methods of translation of different orders of abstractions—dynamic into static, and vice versa—which is a neurological structural necessity for being rational and sane. And surely science should try to be rational. It should be stressed again that in our problem numerical values matter very little, but structure and method, for the many reasons already explained, are of paramount importance. Perhaps even the value of numbers is due mainly to the structural fact that it has forced upon us extensional and relational methods. It is the only language which is in accordance with the structure and functioning of the nervous system, and so helps to co-ordinate these activities instead of disorganizing them.

That these simple structural dependences have been discovered so late is really astonishing. The only explanation I can give of this is that we have been so engrossed in generalizing and generalizations that we lost sight of the fact that in life we deal structurally with *absolute individuals*, and that the only *language* which preserves the *extensional structural individuality* for its elements is found in mathematics—specifically, in numbers.

It may be that a study of mathematical structure and the psycho-logics of mathematics will give results of unparalleled human values, particularly for our sanity. The problems of sanity are problems of adjustment, and no means of adjustment should be disregarded. It may also be that the main importance of mathematics will be found some day to be more in the mathematical *methods* and structure which it has originated, methods forced upon the mathematician

by the relational character of the entities he has to deal with, than in the possible combinations of these entities themselves.

At any rate, we must sadly admit that the problems of mathematical methods and structure and the psycho-logical values of mathematics have so far received very little attention, since we have failed to realize their human importance. In the future this problem will be further, and thoroughly, investigated.

PART IX

ON THE SIMILARITY OF EMPIRICAL AND VERBAL STRUCTURES

The theory of relativity has resulted from a combination of the three elements which were called for in a reconstruction of physics: first, delicate experiment; secondly, logical analysis; and thirdly, epistemological considerations. (457) BERTRAND RUSSELL

The essence of Einstein's generalization is its final disentanglement of that part of any physical event which is contributed by the observer from that which is inherent in the nature of things and independent of all observers. (21) E. T. BELL

Even Leibniz formulated the postulate of continuity, of infinitely near action, as a general principle, and could not, for this reason, become reconciled to Newton's Law of Gravitation, which entails action at a distance and which corresponds fully to that of Coulomb. (547) HERMANN WEYL

This limitation to what is directly observable is ultimately based on Mach's philosophy and, directly inspired by Mach, led three decades ago to the propagation of the so-called theory of "Energetics," which sought to recognise only quantities of energy as physically given and observable quantities. (481) A. SOMMERFELD

CHAPTER XXXV

ACTION BY CONTACT

> The difficulty involved is that the proper and adequate means of describing changes in continuous deformable bodies is the method of *differential equations*. . . . They express mathematically the physical conception of contiguous action. (45)
>
> MAX BORN

The analysis of 'matter', 'space', and 'time' from the point of view of structure and of orders of abstractions has led us to far-reaching conclusions. Let us summarize the semantic results, and consider some of the immediate consequences.

We may begin by recalling the difference between the lower order and higher order abstractions. The lower order abstractions are given to us by the lower nerve centres. They are 'dynamic', 'continuous', non-permanent, shifting, unreliable, and above all *un-speakable.*

They have a character of immediacy, because, structurally in terms of order, they are closest to outside events. They come first in order in the functioning of the nervous system. We always associate with them some 'objectivity' as, by necessity, the eventual definition of an 'object' starts at this level.

It should be emphasized over and over again that, speaking correctly, *on this level* we cannot define anything, since abstractions on this level are fundamentally *un-speakable.* We may look, listen, handle, feel . , but *cannot speak* and therefore cannot define. The moment we *define* our objects, we are no longer on the level of lower order abstractions. By neurological structural necessity we have passed to the higher nerve centres (speech), and higher order abstractions. This is what is meant when we say that this lower level is *un-speakable.*

Because these lower order abstractions are closer to the outside events, and because they come *first in order*, they have a special character of immediacy, with which we *must start.* The struggle begins when, through some primitive-made doctrines or structural assumptions (metaphysics), we try to avoid going any further than these lower order abstractions. As a matter of fact, this is an impossibility, because of the very structure of our nervous system. However intensely we believe that it is possible to do so, and however 'emotionally' we attempt to do it, we are cherishing delusions, which easily become *morbid* identifications, delusions, illusions, and often hallucinations.

This level being *un-speakable*, the only way to function on this level is to look, listen . , but to be *silent* outwardly as well as *inwardly.* This last condition represents a most beneficial semantic state, really difficult, perhaps impossible, to acquire without training.

The higher order abstractions appear to be products of the activity of the higher nerve centres, further removed from the external events and lacking, therefore, in immediacy. But these higher abstractions are static and so may

be analysed. They have a separable unit 'quantum' structure, which can be treated individually. It should be carefully noticed that the static character of these higher order abstractions is the origin of their separable quantum character, conditioned by the human nervous structure. They are, if properly treated, reliable and are uniquely responsible for our being time-binders.

Again, by the structural necessity of our nervous system, we deal first with lower order abstractions, and next with higher order abstractions. It must be noticed that *no one*, unless he is (pathologically) entirely deprived of the higher nerve centres, is, or can be, an exception. We all deal with lower abstractions first, and with the higher next, no matter how perfect or imperfect these abstractions may be.

The general confusion of orders of abstractions, the lack of theories, and therefore of structural understanding of the entirely different characters of these distinct orders of abstractions, leads to, and must result in, identification or confusion of orders of abstractions. As the different processes are going on, whether we will it or not, in every single one of us, they may result in the *delusional* ascribing of the characteristics of the higher order abstractions to the lower order abstractions, as for instance, permanence, immutability . , somewhere involving 'infinities'. When objectified, we have such semantic disturbances as fanaticism, absolutism, dogmatism, finalism . , which often become morbid semantic states.

A similar confusion may lead to the delusional ascribing of the characteristics of the lower order abstractions to the higher ones. Under such delusions we ascribe to the higher abstractions fluidity, shiftiness, non-permanence, 'non-knowability'. , which results in pessimism, cynicism, disregard for science, bitterness, fright, hopelessness and other equally vicious semantic disturbances. These in turn affect by structural necessity the proper working of the entire organism, which always works as-a-whole.

The *A*-system and other older systems were not only built before these facts became structurally known but were actually based on such confusion. Hence their viciousness. By building a language and a method of this nature they perpetuated and made effective *mechanically* through the structure of language, a harmful confusion. This language being not in accordance with the structure and functioning of the nervous system and the world, must produce pathological results somewhere.

We have already seen that the use of the 'is' of identity is unconditionally delusional. Naturally, attitudes (affective, lower order abstractions) which can assert, (higher order abstractions) that so and so on objective levels 'is' so and so, must lead to pathological results. In science this is a profoundly unsatisfactory state of affairs and needs structural revision.

Mathematicians, though in the main unconscious or innocent of the structural, semantic, and neurological issues involved, nevertheless have solved this problem by producing methods of passing from one order of abstractions to another, from dynamic to static, and vice versa. The influence of these discoveries has also affected the other sciences *unconsciously*. Without consciously

recognizing it, the modern trend of science is to banish from its habits and methods the application of the 'is' of identity.

So in science we have to use an actional, 'behaviouristic', 'functional', 'operational' language, in which we do *not* say that this and this 'is' so and so, but where we describe *extensionally* what happens in certain *order*. We describe how something *behaves*, what something *does*, what we *do* in our research work , . If one asks, for instance, what *is* 'length', what *is* 'space', what *is* 'time', what *is* 'matter'. , the only correct answer would be, 'As you asked the question verbally, and I answer it *verbally*, the above *terms remain terms*, which beside structure, have no connection whatsoever with the external world'. Yet undoubtedly we are interested in this external world and we should like to use a language which would help us in understanding this world better. What shall we do? It seems that if we produce a language which is *similar in structure*, to the external world, somehow, as a map or picture is similar in structure to the region it pictures, we should have a uniquely appropriate language. How can we do it? It is quite simple the moment we discover the principle. First of all, abandon completely the *A* 'is' of identity, and, instead, describe ordered happenings in an actional and functional language. Such a language shares with the external world at least the multi-dimensional order of happenings, and it gives us a solution.

It is easy to see that arguments (verbal) about 'matter', 'space', 'time'. , will never become anything else than verbal. All uses of the 'is' of identity, must lead to delusional evaluation. The situation is radically changed when we use an actional or functional language, when we describe what a physicist does when he finds his 'length' or 'second' or any other entity he is interested in.

We should notice here that the above procedure involves extremely far-reaching structural and semantic consequences. First of all, we abandon the vicious use of the 'is' of identity, and eliminate the semantic disturbance called identification. We introduce automatically the full psycho-logical working mechanism of *order*, *extensional* methods and discrimination between the orders of abstractions. We introduce the four-dimensional and differential methods, we build up static units, 'quanta', and so introduce *measurement* and its language called mathematics, which leads to structure and so to knowledge at each date.

It will be useful to recall why mathematics and measurements are somehow so important in our lives. Our nervous system, as we have seen, exhibits different activities on different levels. On one level the abstractions are shifting, non-permanent; on the other static and permanent in principle. This is expressed in our lives in a longing for some permanency, some security, some 'absolutes'. Mathematics formulated this tendency first and with *full success*. Mathematics has not only formulated full and successful theories of 'change', as, for instance, the theory of functions and the different calculi, but also full-fledged and remarkable theories of *invariance* under transformations. These new theories of invariance are actually *absolute* formulations in the only sense in which the

term 'absolute' has a meaning; namely, relative, no matter to what; all of which leads to the only content of knowledge—structure.

The whole Einstein theory should, in this sense, be called the 'theory of the absolute', and can be expressed as the simple demand that 'universal laws' should be formulated in an invariant form, a most revolutionary demand and yet so *structurally natural* that no one can deny it.

When we mathematize or speak about potential or actual measurements, we are dealing with *ordered*, *extensional*, actional, behaviouristic, functional and operational entities, and so we build up a language which at least has a similar *structure* to the external events. Numbers imply units, quanta, but also order. It seems that number is the only abstraction upon which we all must agree. We never doubt that a statement, such as that 'I have in my pocket five pennies', may be perfectly definite and ascertainable for all. The specific and unique relations called numbers seems to have absolute significance. It must be added that the existence of non-quantitative branches of mathematics does not alter what is said here. In these branches, the asymmetrical relation of order remains paramount and we may treat numbers from either of their two aspects, the cardinal or the ordinal.

The epoch-making significance of the Einstein-Minkowski work consists precisely in the fact that they were the first to *apply* the above, though without, it is true, formulating the general principle. The lack of such a general, $\bar{A}$, epistemological formulation retards considerably the understanding of their work, and so laymen miss the enormous structural, and semantic beneficial effect upon the proper working of our nervous system and our sanity.

Before giving a short methodological account of the Einstein theory it will be well to recall some structural and semantic conclusions which the differential calculus suggests.

When we were dealing with the notion of a variable, we saw that the variable might be *any* element selected out of an ordered aggregate of elements. We can select elements relatively widely separated from each other, as, for instance, the numbers 1 and 2, or points, let us say, an inch apart. It is obvious that if we choose, we can make the gaps smaller, and postulate an infinity of intermediate steps. When we make our gaps smaller, the elements are ordered more densely and closer together. In the limit, if we choose indefinitely many elements between any two elements, our series become compact, if we still have a possibility of gaps; or they eventually become what we call *continuous*, when there are no more gaps.

Without legislating as to whether the entities we use in physics are 'continuous', 'compact', or 'discontinuous', we may grant that the maximum elucidation of the above terms in mathematics is very useful. We can easily see that in terms of *action* a continuous series gives us *action by contact*, since consecutive elements are indefinitely near each other. As the differential and integral calculus were built on the structural assumption of *continuity*, the use of the calculus brings us in touch not only with our x but also with its indefinitely close neighbour $x+dx$. We see that the calculus introduces a most important

structural and semantic innovation; namely, that it is a language for describing *action by contact*, in sharp contradistinction to the structural assumption of action at a distance.

Let us illustrate the above by a structural example. Consider a series of equal small material spheres connected with each other by small spiral springs as shown on Fig. 1.

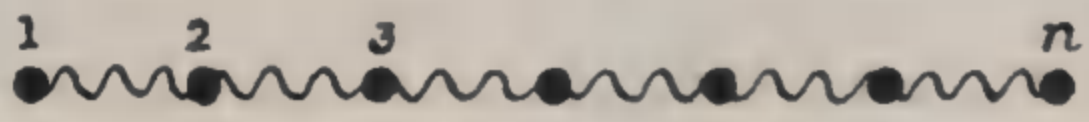

FIG. 1

These little spheres all have inertia, because of which, and because of the little springs, they resist displacement. If we displace the first of our spheres either in the transverse or longitudinal direction, it acts upon the second sphere, which in turn acts upon the third , . We see that the disturbance of equilibrium of the first little sphere is transmitted like a wave to the next sphere and so along the whole series. The most significant point in the analysis of such a wave of excitation is that it is not transmitted with some 'infinite velocity', or 'infinitely quickly' or in 'no time'. The action of each sphere is slightly delayed owing to its inertia, that is, it does not respond 'instantaneously' to an impulse. It must be noticed that the displacement is not due to a velocity, but to an acceleration, which is a change of velocity and requires a short interval of 'time'. The change in velocity again requires an interval of 'time' to overcome inertia and produce displacement. Similar reasoning applies to a long train just being started by the engine. The cars being coupled together by more or less elastic means, the engine may be moving uniformly and some of the last cars still be stationary. The pull of the engine is *not* transmitted instantaneously but with a *finite velocity*, due again to the inertia of the cars.

We see that the only structurally adequate means of describing changes in continuous, deformable materials is to be found in differential equations which express a method of dealing with *action by contact.*

We have already seen that this action by contact involves also the *finite velocity* of propagation, a fact of crucial structural and semantic importance. In the history of science we can distinguish three periods. The first was naturally the period of action at a distance, the best exemplified by the work of two great men, Euclid and Newton. In it we find of course, a superabundance of 'infinities'. With the advent of the differential calculus, and the introduction of differential equations in the study of nature, the notion of action at a distance became more and more untenable. We had a period of pseudo-contiguous action, which indeed involved differential equations; but the *velocity of propagation* was not introduced explicitly, and so there remained an implicit structural assumption of 'infinite velocity' of propagation. As an example of such pseudo-contiguous action we can cite the older theories of potential, which give differential equations for the change in the intensity of the field from place to place,

but which do not contain members that express a change in 'time', and hence do not take into account the transmission of electricity with finite velocity.[1]

The modern theories, as for instance, the Maxwell theory of electromagnetism, and the Einstein theory, are based on *action by contact.* These theories not only use the differential method, but they also introduce explicitly the *finite velocity* of propagation.

The invention of differential geometry with the recent contribution of Weyl, which we have already mentioned, transforms the geometry of Euclid from a language of action at a distance into contact geometry, or a language of indefinitely near action.

It should be mentioned perhaps that the riemannian differential geometry is more general than all the $\bar{E}$ geometries which preceded it, and includes them, as well as the E geometry, as special cases. Perhaps, as Weyl points out,[2] the investigation of the famous fifth postulate, which was the beginning of $\bar{E}$ geometry, was accidental in importance and the main structural value of the $\bar{E}$ geometries lies precisely in the application of the differential methods to geometry which was originated by the great work of Riemann. This work, we see, has carried us from metaphysical action at a distance to a physical action by contact. In passing from the older mechanics to electromagnetic events a very striking analogy appears, which explains the finite velocity of propagation.

In mechanics, when we have waves in an elastic medium, the finite velocity of propagation is due to the delay which occurs due to the inertia of materials. Now inertia is determined by acceleration (d^2s/dt^2), which represents the rate of change of the velocity ($v=ds/dt$), velocity itself being a rate of change of displacement. We see that this retardation, or negative acceleration, is represented by a double differentiation.

Something analogous occurs in electromagnetic events. The rate of change of the electric field (de/dt) determines the magnetic field; and then the rate of change (dh/dt) of the latter determines the electric field at a neighbouring point. The advance of the electric field from point to point is thus conditioned by two differentiations with respect to 'time', which is quite analogous to acceleration.

It is due to this double differentiation with respect to 'time' that the formulation of electromagnetic waves are structurally possible. If the partial effects were to occur without loss of 'time', no propagation of the electric waves would occur. The maxwellian 'field equations' not only express the above-mentioned relations, but introduce structurally the finite velocity of propagation which makes the Maxwell's electromagnetic theory structurally a contact theory.

The Einstein theory is also structurally a contact theory, and it may be said that it was originated by this contact tendency, and has carried it to the limit, as we shall see later. The gaussian theory of surfaces, whose extension to any number of dimensions was made by Riemann, also represents action by contact. This theory does not state the laws of surfaces on a large scale,

but only their differential properties, the coefficients of the measure determination, the invariants which we can form, and the curvature and its measure. The form of a surface and its characteristics can then be calculated by a process similar to the solution of differential equations in physics.

We are now in a position to understand why the newer physics and the $\bar{N}$ systems, which are built entirely on the foundations of action by contact, found the E-system unsatisfactory. The E-system was built on the structural assumption of action at a distance, and we had to select the $\bar{E}$ geometries as originated by Gauss, Lobatchevski, Riemann, and others, which gave to physics the necessary geometry of action by contact.

But the question of action at a distance versus action by contact has also an experimental aspect which makes the latter theory more satisfactory.

Faraday (1791-1867) was not a learned academician, and he was much freer from scientific prejudices than any of his contemporaries. From a bookbinder's apprentice he became through his genius one of the founders of modern physics. His method of experimenting was to try every possible experiment and note what happened.

In 1838 Faraday made an important structural discovery; namely, that the mutual action between two electrically charged bodies depends upon the character of the intervening medium. Faraday established by this experiment that the capacity of a spherical condenser changes when another material is used as the separating medium, rather than air. He found that the capacity became twice as large when the medium was paraffin, three times as large for shellac, six times as large for glass, and about eighty times as large for water.

This experiment became the foundation of the new theory. The old 'action at a distance' theory postulated that the electrostatic field was merely a geometrical structure without physical significance, while this new experiment showed that the field had physical significance. Every charge acts first upon its immediate surroundings, and it is only through the medium of these that the action is propagated. The discovery of displacement currents necessitated an extension of his point of view to all distances.[3]

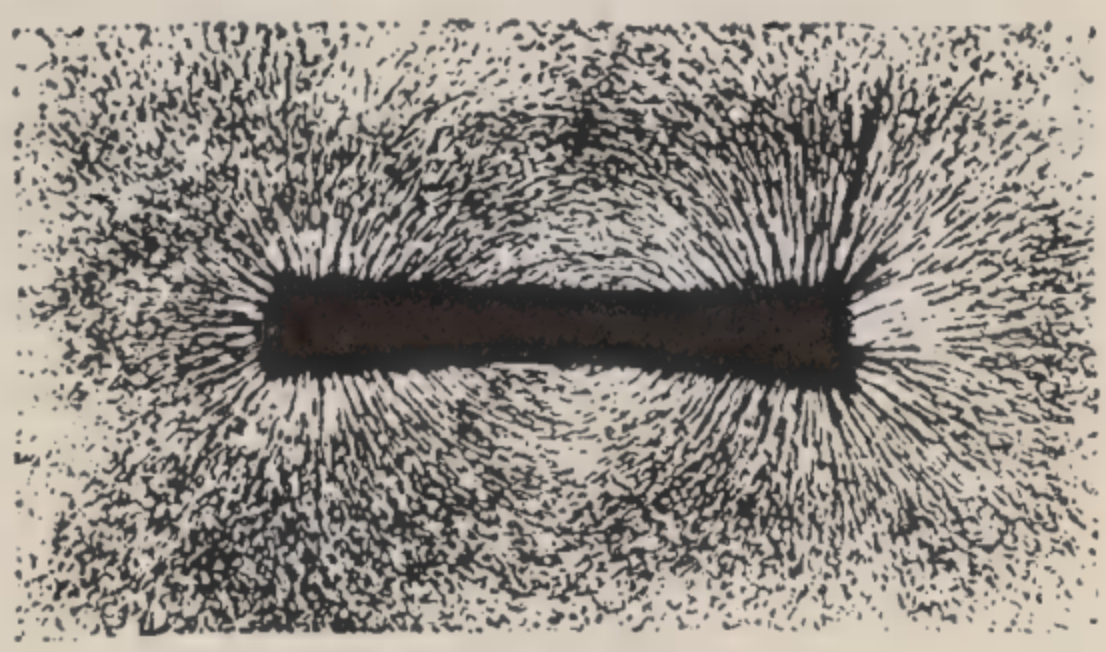

FIG. 2

Faraday was so impressed by this discovery that he abandoned the older theories of action at a distance and formulated a structurally new theory of contiguous action for electric and magnetic events. Any one can convince himself of the fact that the fields represent very actual physical conditions by taking a sheet of paper, sprinkling some iron dust upon it and

putting a magnet under the paper. He will find that the particles of iron dust arrange themselves in a very definite structure as shown in Fig. 2.

Faraday also discovered that the forces between two magnetic poles likewise depend on the medium that happens to be between them. He concluded correctly that the electric as well as the magnetic forces are produced by a state of tension in the intervening medium.

These two examples will suffice as illustrations, but it can be said in general that all modern physics gives ample proofs of the correctness of Faraday's structural point of view. Some physicists, for instance Helmholtz, built special devices to test the correctness of this theory. As a matter of fact the success of the whole electromagnetic theory of Maxwell, which is structurally built as a contact theory, in which the velocity of propagation is considered finite, is in itself one of the best proofs of the correctness of the theory.

The finite velocity of light was discovered by Olaf Römer in 1676 and has since been repeatedly verified. This velocity is usually denoted by c and is known to be approximately 300,000 kilometres per second, that is, $c = 3 \times 10^{10}$cm./sec., or 186,000 miles per second.

In 1856 Weber and Kohlraush calculated a certain constant which appears in the electromagnetic theory, and discovered that the constant had the dimension of velocity, $[c] = \left[\frac{L}{T}\right]$, and that its numerical value was 3×10^{10}cm./sec., which is the exact value for the velocity of light. This fact led Maxwell to associate light with electromagnetic waves, a view justified by experiments. In 1888 Hertz not only established once more the interrelation of optics and electrodynamics but found that the velocity of propagation of electromagnetic waves is finite and exactly equal to the velocity of light.[4]

Outside the exact sciences the principle of action by contact is making but slow progress, perhaps because of A *s.r* and the lack of structural formulations of the general issues at hand. We are happy to find a notable exception in the biological work of Professor C. M. Child, who has laid down a foundation for $\bar{A}$ biology and his system is structurally based on action by contact. This $\bar{A}$ biology has been applied to neurology by Professor C. J. Herrick. This present work, being a $\bar{A}$-system, must follow the methodological and structural advances explained here, and the $\bar{A}$ biology and neurology founded by Child and Herrick.

It is interesting to follow up the structural merging of geometry and physics. There are certain Smiths and Browns who call themselves physicists. There are some rooms with various instruments, which are called physical laboratories. The activities of the physicists which interest us are twofold. First, these scientists come to their laboratories, manipulate their instruments, note the positions of some pointers, manipulate the instruments again, note positions again , . This represents the *un-speakable* level of activity. Whatever happens happens, but there is no speaking to be done on that level.

Later the scientist describes his experiments in words. Obviously there are two entirely different stages in building physics, which usually we do not distinguish.

Quite obviously the *un-speakable* level cannot be called 'physics', and so we must apply the term to the higher order abstractions on the *verbal level;* namely, to the reasoned verbal account of what the experimenter saw, or felt, or experienced, in general abstracted on the lower levels; summarized, generalized . , in higher orders.

Physics represents then a verbal discipline. Being verbal, it needs a language. What language shall we select? As we want to have a science called physics, we shall naturally try to use the most structurally correct language in existence, so by necessity we must look in the direction of mathematics.

In mathematics we find originally two entirely different disciplines. One we may call arithmetic; the other, geometry. Becoming acquainted with these two originally separated languages, we find that the actual experiments and the stimuli for many experiences of importance to us, are outside our skins; so we try to choose the one of these two languages which is the more closely related in structure to the lower abstractions—that is, to what we see, feel , . Naturally we have an inclination toward the geometrical languages, dealing with 'lines', 'surfaces', 'volumes' . , terms for which we find immediate and quite obvious applications. By further investigation we find that of late both languages have become so developed in structure, that either can be translated perfectly into the other. This fact makes geometry the link between the higher order abstractions and the lower order abstractions. We have seen that *physics,* as well as geometry, must be considered *verbal* disciplines and their fusion becomes a very natural fact.

It is true that, as yet, 'time' appears as the bothersome factor, but 'time' may very well be represented geometrically, except that our diagrams and figures look a little different. For instance, a *flat* circular orbit in two-dimensional 'space' becomes a helix in three-dimensional space-time, a vibrational motion in one-dimensional 'space' becomes a wave-line in two-dimensional space-time , . 'Time', when properly represented, becomes simply another geometrical dimension.

It should not be forgotten that mathematicians obtain most of their structural inspirations from physics and build up mathematical theories to supply the structural needs of the physicist. We see an excellent example of this in the $\bar{E}$ geometries. In the days of Euclid, when physics hardly existed, we had 'emptiness', 'action at a distance', and such notions as were quite satisfactory for the needs of surveyors and builders. With the development of astronomy and physics, curved lines became more and more structurally important, and the haziness of the definition of 'straight line' also became apparent. The notion of 'emptiness' also became slowly structurally untenable. Such geometers as Gauss, Lobatchevski, and others, began to demand that the axioms of geometry be tested by experiment. With the introduction of 'curvature', the 'straight line' became only a special case of a curve with zero curvature.

The invention of the differential calculus also had a tremendous structural influence. It introduced continuity as a basic assumption in the vast structure of science, and cleared the way for psycho-logically trained scientific workers in structural continuity, and therefore in *action by contact.*

The discovery that light appears as electromagnetic waves and the finite velocity of both, made the notion of 'absolute emptiness' structurally untenable; and so E geometry with its action at a distance, 'emptiness' and neglect of gravitation and electricity, became very unsatisfactory. Indeed if our universe were E, light could not reach us.

Leibnitz, who invented the differential calculus independently of Newton, formulated a postulate of action by contact, and therefore could not become reconciled to Newton's Law of Gravitation which was structurally a law of action at a distance, corresponding fully to Coulomb's law in electricity. The latter law states that the force exerted by two electrically charged bodies upon each other is inversely proportional to the square of the distance between them, and acts in the direction of the line joining them.[5]

The introduction by Faraday of the structural notion of a 'field', instead of the notion of electrical charges acting at a distance, introduced the notion of a strain of the electrical field, which appears structurally as 'lines of forces'. Here we already have a 'fulness' of 'lines' and a big step toward the structural fusion of physics with geometry has been taken.

The transition from E to riemannian geometry corresponds structurally to the transition from physics based on action at a distance to physics based on action by contact. The fundamental metrical theorem of E geometry is the pythagorean rule, which expresses the fact that the square of the distance between two points is a quadratic form of the co-ordinates of the points. If we regard this theorem as strictly valid only in the case of points which are very near each other, we pass at once from E geometry to differential geometry. By doing so we gain a notable structural advantage, as we dispense with the necessity of defining our co-ordinates more precisely; because the pythagorean law, when expressed in differential form, is invariant for arbitrary transformations.[6]

Semantically, Riemann was the immediate predecessor of Einstein, although Einstein was not directly influenced by him. In differential geometry we ought to start with indefinitesimally near points, and depend for the analysis of greater distances, areas and volumes, on integration. The difficult notion of 'straight line' has to be replaced by the notion of the shortest line (geodetic), which is easily defined by differential methods and found empirically. In the older method, the length of a curve was to be found, in general, by the process of integration. The length of a 'straight line' between two points was supposed to be defined as a whole, and not as the limit of a sum of indefinitely little bits. Riemann considered that a 'straight line' does not differ in this respect from a curve. Measurements which are always performed by means of some instrument are *physical* operations, and their results depend for their interpretation

upon the theories of physics. Dealing with geodetics is therefore preferable to dealing with 'straight lines'.[7]

We see that the problem was ripe for a final stroke of genius. Einstein's structural discovery of the dependence of 'space' and 'time', and Minkowski's success in giving a geometrical interpretation to the Einstein theory accomplished the probably irreversible fusion.

Three-dimensional kinematics becomes four-dimensional geometry, *three-dimensional dynamics* can be considered as *four-dimensional statics.*

We see immediately the human, psycho-logical, semantic and neurological importance of this fact. Our nervous system by its structure produces abstractions of different orders, dynamic on some levels, static on others. The problems of sanity and adjustment become problems of translation from one level to another, for which the structural advances in science supply us with methods of solution.

It should be noticed that the semantic gain due to the above facts is considerable, and that being structural, it is practical as well as theoretical. The fact that geometry has lost its old restricted status, which formerly applied principally to what could be 'intuitively visualized' and has been further abstracted to apply to what can be 'conceived', has merged geometry with the rest of mathematics. This merging represents a great structural and semantic step forward, and makes possible the treatment of geometrical problems by purely analytical means. It liberates geometry from the restrictions of lower order abstractions. By using 'geometrical intuition' (lower order abstractions) we find again a great help in analysis.

In the cyclic nerve currents, our so-called 'intuitions' (lower order abstractions) are not structurally isolated from our 'conceptions' (higher order abstractions), but both are intimately connected and influence each other. Modern advances are not only in perfect accord with the 'organism-as-a-whole' principle, but indeed give us excellent proofs that this principle is sound. 'Psychologists' miss a great deal by disregarding this important and unique form of human behaviour which we call mathematizing.

CHAPTER XXXVI

ON THE SEMANTICS OF THE EINSTEIN THEORY

It is precisely here, in an improved understanding of our mental relations to nature, that the permanent contribution of relativity is to be found. We should now make it our business to understand so thoroughly the character of our permanent mental relations to nature that another change in our attitude such as that due to Einstein, shall be forever impossible.
(55) P. W. BRIDGMAN

It is not my aim to expound the Einstein Theory as such. There are many excellent and competent books written on this subject. I have already explained and stressed several structural points which in the last analysis are the foundation of Einstein's work. Many 'thinkers' through the ages have felt vaguely the dangers of the structure of language and the viciousness of objectification, that is, of the delusional ascribing of objective values to verbal forms. This vague feeling, of course, is useful in individuals, but it is a private benefit, which cannot be made public without some sort of formulation. The stroke of genius of Einstein was that he produced a *non-elementalistic*, linguistic system of new structure. Einstein, being a physicist, decided rightly, as we understand now, to be entirely actional, behaviouristic, functional, and operational, and to stop gambling on words. The older *el* linguistic problems of 'matter', 'space', and 'time' were in such a mess, due to the objectification of verbal structures, that it was useless to talk any more in the old way. He decided to describe what a physicist *does* when he measures 'space' and 'time', and to abandon, perhaps unconsciously, the 'is' of identity.

It seems unnecessary to stress the simple fact that when we measure a piece of wood, for example, we mark it off with another piece of material which we have accepted arbitrarily as our 'unit of length'. The coincidence of our 'unit' with the intervals between the marks is again judged by an extremely complex electromagnetic-neural process, which was quite disregarded until Einstein. Our judgement is conditioned by the light rays travelling with finite velocity which excite our nervous system through the retina, this excitation in turn also travelling with finite velocity. We see that the apparently simple measurement of a 'length' is really an extremely complex process, in which the *finite* velocity of light and of the nerve currents plays a very important role. Naturally, if we were to assume an 'infinite' velocity of the propagation of light, our verbal speculations about 'space' and 'time' might be perhaps entertaining, but they would nevertheless be fundamentally and structurally wrong.

Similar remarks apply to the measurement of 'time'. What do we mean when we say that a train has arrived at the station at 9 o'clock? We mean no more and no less than that the arrival of the train coincided with the arrival of the pointer of a clock at a point marked 9 on the clock face. In other words,

we saw 'simultaneously' the arrival of the train and the pointer of the clock reaching the mark 9.

Our judgement about the results of measurements of 'time' depends on the *seen* coincidence of events—in this case, of the arrival of the train with the arrival of the pointer of the clock at the mark 9. Similar considerations, which applied to the measurements of 'lengths', apply also to the measurements of 'time'.

We see with Einstein that if we want to make any headway we shall have to investigate the two key terms; namely, 'velocity' and 'simultaneity'.

The newtonians take a particular delight in accusing Einstein of being a 'psychologist' and not a physicist. We have already stressed the physical subjectivity of physical instruments. What is said there applies, not only to the retina of the eye, but also to a photographic camera, or to a microscope or telescope, or any other instrument. Before an energetic packet, be it a light-impulse or a bullet, is able to accomplish any result it must first reach its mark, and so the finite velocity of propagation must be taken into consideration, which is a hard, established, empirical structural fact. So the criticisms of the newtonians are simply shallow and unscientific (1933). They disregard most important empirical physical facts, and so simply defend a semantic disturbance without aiding science (1933).

With the einsteinians, we treat the eye on the same footing as we would treat the camera or any other physical instrument. Even the newtonians must admit that when they photograph some happening on the sun, for example, the happening actually occurred (approximately) eight minutes *before* the photographic plate was affected. The eight minutes is the 'time' taken by the light to reach the earth from the sun.

Let us analyse the term 'velocity' first. We find ourselves here, as in any other human problem, on two distinct levels of abstraction, and we must discriminate between them.

Let us take up the verbal level first. We see that before we can talk about our terms 'space' or 'time', 'length' or 'seconds', we have to know a great deal about the term 'velocity'. How do we define the term 'velocity'? We define it as 'space divided by time', $v = s/t$. We see that on the *verbal* level the situation is perfectly hopeless and no result can be expected from verbal gambling. It may be added that older notions were based on objectification, or confusion between the two levels of abstraction, and the affective belief in the magic of words, identification playing most of the structural havoc.

How about the instrumental level, the silent level of the lower order abstraction? On this level, we find that physicists in their actions, behaviour, operations . , have elaborated a fairly definite technique for finding the data they require. So we see that there is no choice, we must *start* on this level.

But starting on this level is not all, and not enough. We must somehow *talk* about these doings and operations. Hence we must select a language which in its *structure* will reflect the structure of these actions and operations. Therefore we must abandon the 'is' of identity and *describe* in the asymmetrical

language of order the happenings recorded by an instrument or by our lower nerve centres.

Without going into details we may summarize the results as achieved by the physicists. Experiments by the physicists, as indicated by the coincidences of pointers on different instruments, have seemingly established the fact that the 'velocity' of light, as defined by *behaviouristic*, *operational* instrumental means, is a *constant*, $c = 3 \cdot 10^{10}$ cm./sec., independent of the relative velocity of the observers. By the 'observers' we mean again the readings on the instruments which the observer carries with him. Now this result contradicts flatly the established *verbal expectations* which we reached on verbal levels through the elementalistic structure of language and the semantic disturbance, of ascribing 'objective' existence, to the *terms* 'space' and 'time'.

The situation is acute. Shall we follow our semantic disturbances and reject hard empirical structural facts, or shall we accept the experimental facts and eliminate semantic disturbances?

As usual, the answer is implied by the method of putting the question. We accept the experimental facts and revise our semantic disturbances. In this case a psychiatrist might be a useful co-worker with the physicist.

The einsteinian revolution is structurally and semantically so fundamental, that every intelligent person should be acquainted with it. It will therefore be as well to consider some of its details.

In classical mechanics we had the classical mechanical principle of relativity; namely, that all mechanical equations have one form for two co-ordinate systems moving uniformly with respect to each other. The above has a very simple empirical meaning. If we travel in a train, let us say at a velocity of 50 miles an hour, all our activities in the train have one familiar relative velocity as if the train were at rest. If we throw a ball with a velocity of 20 miles an hour to another passenger on the train in the direction of the movement of the train, the ball will not reach the other passenger with the velocity of 20 miles an hour *plus* the additional 50 miles an hour velocity of the train but will reach him with the velocity as if the train were standing still. Not so, however, if the ball were thrown to an observer, standing on the tracks. The ball might hurt him, because it would have, relative to him, the velocity of 20 miles an hour of the ball, plus the velocity of 50 miles an hour of the train, or in all, a velocity of 70 miles an hour.

Quite probably, even our remote ancestors who used artificial means of transportation on land or water did not overlook the structural fact that mechanical events happen in just one way, whether the system is at rest or in relative motion. With the advent of *verbal* formulations of physics and mechanics, such happenings were formulated verbally; and so, slowly, the language of old structure with its consequent objectifications was built.

Now on *verbal grounds*, which seemed to be justified by experimental, macro-mechanical facts, we concluded that one law should prove valid in the case of electrodynamic and optical events.

To reformulate the above in simple symbols, let us imagine two parallel co-ordinate systems, O' and O'', of which the second moves with a velocity u relative to the first in the common x direction. If we denote the co-ordinates of the first system by single primes, and the co-ordinates of the second system by double primes, then, as usual, the co-ordinates of a point, P, in the second system would be connected with its co-ordinates in the first system by the equation $x''=x'-ut$, which means that the x'' co-ordinate is less than the x' co-ordinate by the amount that our second co-ordinate system has moved; namely, by $a=ut$. We gave the diagram in two dimensions because it is simpler,

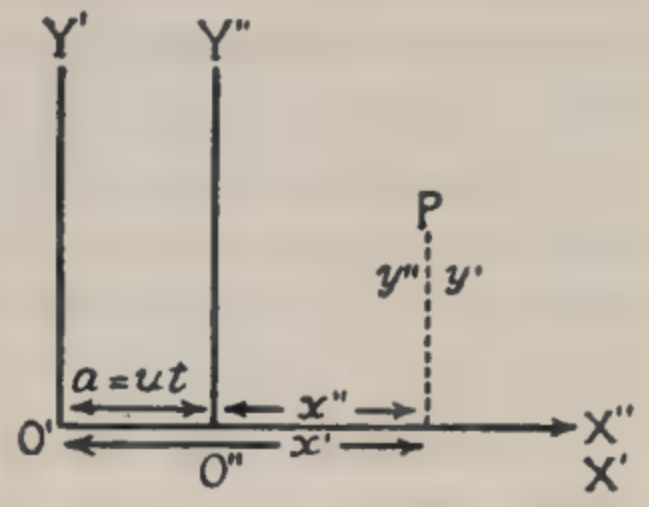

FIG. 1

and, as we have assumed that the displacement is parallel to the x axis, the other co-ordinates remain unaltered, $y''=y'$, $z''=z'$. 'Time' by the older assumptions, being 'objective' and 'absolute' would be 'the same'; namely, $t''=t'$ ('absolute time'). The classical law of relative motion states that if the equation of motion in the first system is $f'(x', y', z', t)=0$, this function must also be zero when x' is replaced by its new value; namely, $(x'-ut)=x''$ so that $f'(x'', y'', z'', t)=0$.

Let us see if the above conditions hold true when we deal with the *propagation of light in spherical waves.*

$$OB^2=OA^2+AB^2$$
$$OB^2=x^2+y^2$$
$$OP^2=OB^2+BP^2$$
$$OP^2=x^2+y^2+z^2$$
$$c^2t^2=x^2+y^2+z^2$$

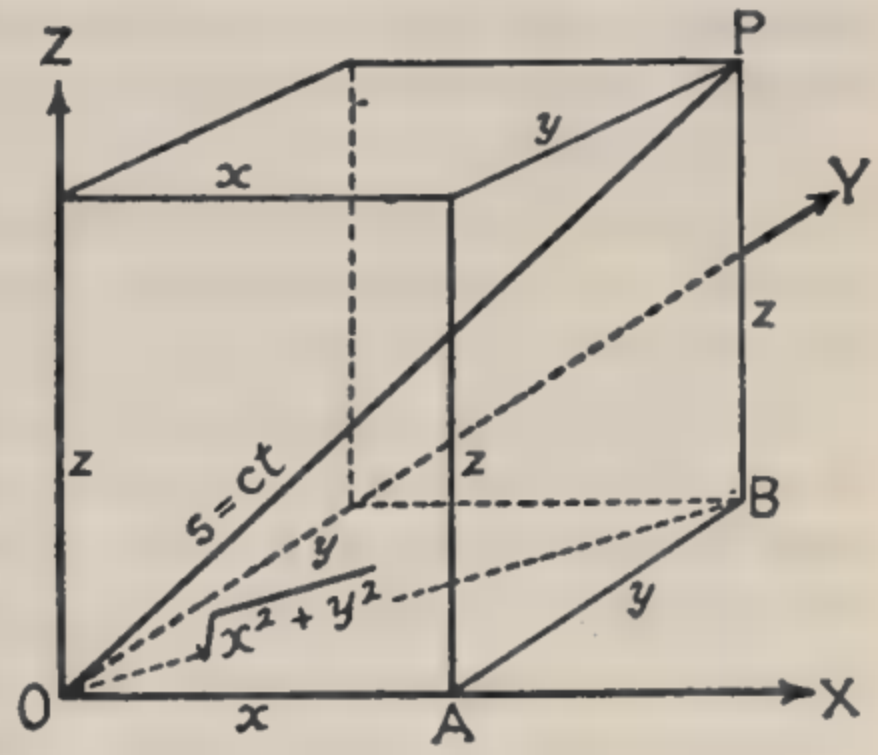

FIG. 2

If we select a three dimensional co-ordinate system O, the distance s of the point P from O is equal, by the pythagorean rule, to $s^2=x^2+y^2+z^2$. If we assume that a light ray is travelling from O to P, the distance s could then be represented by the product of the velocity of light c by the 'time' or ct. The square of this distance would then be $c^2t^2=s^2$. We have $x^2+y^2+z^2=c^2t^2$, or $x^2+y^2+z^2-c^2t^2=0$.

We can easily convince ourselves that if the last equation for light-waves holds good in the first co-ordinate system it cannot hold in the second.

Writing the last equation in our primed letters, we have $x'^2+y'^2+z'^2-c^2t^2=0$. If we pass to our second system of co-ordinates moving uniformly in the X direction with the velocity u relative to the first system, our y', z', t, do not alter by assumption, but only $x''=x'-ut$. We would have by substituting $x'-ut$ for x'', and retaining the primed values for y'', z'', t:

$$(x'-ut)^2+y'^2+z'^2-c^2t^2=x'^2-2x'ut+u^2t^2+y'^2+z'^2-c^2t^2 \qquad (1)$$
$$=x'^2+y'^2+z'^2-c^2t^2+(u^2t^2-2x'ut).$$

But, by assumption, $x'^2+y'^2+z'^2-c^2t^2=0$, and therefore equation (1) cannot be zero unless $(u^2t^2-2x'ut)=0$. This last condition would mean that our second system of co-ordinates is also at rest. We see that for *light-waves* the older mechanical principle of relativity does *not* hold, as the equations *are altered* when we pass from one system of co-ordinates to another which moves with uniform velocity relative to the first.

To indicate this more obviously, we will express it in formulae. Consider two co-ordinate systems O' and O'', in which the second moves with a uniform velocity in the X direction relative to the first. If for the *light-waves* the equation $x'^2+y'^2+z'^2-c^2t^2=0$ holds in the first system, a similar equation for the second moving co-ordinate system, $x''^2+y''^2+z''^2-c^2t^2=0$, cannot be true. In other words, $x'^2+y'^2+z'^2-c^2t^2 \neq x''^2+y''^2+z''^2-c^2t^2$; whence we have an *inequality*, fundamentally contradicting the classical principle of relativity.

This extraordinary and unexpected inequality, because it contradicted structurally the classical mechanical principles of relativity, which apparently had been well established experimentally, created a baffling semantic situation which was profoundly unsatisfactory.

What could we do about it? Should we abandon the older principle of mechanical relativity; or should we have two different laws, one for the older gross macroscopic mechanical relativity, and another for optical and electrodynamic events; or should we investigate the fundamental structural assumptions which underlie our formulae, and see if the discrepancy is not due to some prejudice or some structural dogma which we have overlooked for centuries?

If a solution of the last kind should be found it would naturally be most satisfactory. The admission of two relativities, one for the mechanical events, the other for the optical events, would be against the whole trend of science, which requires the unification of theories.

Such a structural revision; namely, the rooting out of the old unjustified dogma which made all the trouble, was the work of Einstein's genius. In this epoch-making discovery he was *assisted* in the beginning by the famous Michelson-Morley experiment, since performed repeatedly with similar results, *seemingly* proving that the velocity of light is a constant no matter what the relative motion of the observer. If we take the equation for the spherical propagation of light-waves $x^2+y^2+z^2-c^2t^2=0$ or $x^2+y^2+z^2=c^2t^2$;

then $$c=\frac{\sqrt{x^2+y^2+z^2}}{t}$$ in one system of co-ordinates,

and $$c=\frac{\sqrt{x'^2+y'^2+z'^2}}{t}$$ in another system of co-ordinates,

are equal, which stated in another form would mean that

$$x^2+y^2+z^2-c^2t^2=x'^2+y'^2+z'^2-c^2t^2.$$

The experiment says this relation is true; the arguments already advanced show it cannot be true. So we have to hunt for some error or compensation.

With Einstein's explanation, the finding of the error is simplicity itself. In the older mechanical relativity our 'space' and 'time' were *objectified*, we

endowed them with objective values of definiteness and rigidity, we dealt with 'absolute space' and with 'absolute time', which was 'unchanging' and 'the same for all'. In the older assumptions our *velocities varied.* If A had a velocity of 5 centimetres per second, for instance, and B was overtaking A with a velocity of 7 centimetres per second, the relative velocity between A and B would be $7-5=2$ cm. per second. The *units of 'space' and 'time' were definite, immutable and did not and could not vary*, which followed directly from the assumptions of an 'objective' 'absolute space' and 'absolute time'.

In the case of light, we came in contact with a velocity which did *not vary* for any observer no matter what his relative motion. The velocity c was found to be constant, so the natural assumption to make is that our *'space' and 'time' vary* for different observers.

In the above equations as they stand, 'absolute time', $t=t$, the 'same' for all observers is assumed, which made such equality impossible. Assuming different 'times' for different observers, t for the first, and t' for the second, such a compensation transforms our inequality into an equality, as demanded both by the experiment and by the theory. Instead of writing

$$x^2+y^2+z^2-c^2t^2=x'^2+y'^2+z'^2-c^2t^2,\ (t=t),$$

which *cannot be true*, we write,

$$x^2+y^2+z^2-c^2t^2=x'^2+y'^2+z'^2-c^2t'^2=0,\ (t\neq t')$$

which can be true. We should notice that in the first equation we have on both sides t, which makes the equation impossible, whereas in the second equation we have on the left-hand side t and on the right-hand side a different t; namely, t'.

The above considerations mean that there is a definite structural discrepancy between the old language and the empirical world, requiring a fundamental structural linguistic revision. This revision has been accomplished, and is known as the Einstein theory. It is not implied that Einstein's work is final, but that it shows clearly the structural errors of the old elementalism to which we can never return.

In other words, in the older mechanics we had definite and permanent 'time' (absolute) and varying relative velocities. Dealing with *light-waves* we find experimentally that the velocity, c, of light does *not vary* with the relative motions of the observers and we must assume a *variable time* to preserve our equations.

An obvious objection can be raised to this: why alter our habitual notions of 'time'? Can we not keep the old *s.r* and find some other method of compensation, less bothersome and less revolutionary? The older physicists and Einstein give a long and convincing list of perfectly sufficient reasons for such a change; yet their arguments always leave us somehow in doubt, with the feeling of a lurking possibility that the old can be preserved.

What has already been said in this work about structure and semantic disturbances and the fact that the *terms* 'matter', 'space', and 'time' *are not objects*, which they cannot be, removes perhaps for good and all, the last doubt as to the revolutionary and epoch-making significance and value of the structural linguistic discoveries of Einstein. On these grounds *alone* the return to

the old is impossible. The old is due to objectification of the structural peculiarities of the old *el* language, and to semantic disturbances, which at the present low level of our development is inevitably the result of *copying lower animals in our 'thinking'*, a *pathological* process for 'man'.

It will be well to explain at this point why I said that the Michelson-Morley experiment only *assisted* Einstein, and only *seemingly* proved the constant velocity of light. Historically, there is no doubt that the beginning of the theory of Einstein was suggested by, and had its physical basis in this experiment. In reality, as the whole of this present work about structure shows, the two issues are quite independent. The fact of the *finite* velocity of light has never been challenged, on the contrary it is becoming more and more solidly established, both empirically and theoretically, simply because an 'infinite velocity' has no meaning.

With the structural results of this present work, and the establishment of the fact of the finite velocity of light, the whole Einstein *theory* has a perfectly solid structural, linguistic foundation (1933). Nevertheless it is extremely gratifying that the latest, very important, and painstaking work of Doctor Roy J. Kennedy seems once more to add fundamental experimental support to the correctness of the Einstein theory.* From the point of view of structure, Einstein merely eliminated some primitive, perhaps even animalistic, remains of objectification which still lingered in the structure of our language of 'matter', 'space', and 'time'. These, being animalistic, were unfit for humans; vitiating not only our daily lives but science as well. (Eddington in *The Mathematical Theory of Relativity*, p. 196, uses the term 'pre-human' in a similar connection.)

It must be recalled that the definition of velocity is connected in a *circular* way with 'space' and 'time'. That is, in the definition of the relation of velocity, $(v=s/t)$, 'space' and 'time', the definition of any one of our three terms depends upon our definition of the other two; whence there are many possible ways of verbal adjustment.

As we saw, the mechanical verbal principle of relativity with which we are all familiar was not structurally able to account satisfactorily for a similar relativity of optical and electrodynamic events. The older formulae of transformations were, as already given, $x'=x-ut$, $y'=y$, $z'=z$, $t=t$. These formulae are called the Galileo transformations in honour of the founder of mechanics; and, as we have seen, structurally they are not general enough.

If we consider the equation $x^2+y^2+z^2-c^2t^2=0$ and $x'^2+y'^2+z'^2-c^2t'^2=0$, we find that the galilean transformations do not satisfy them. Lorentz and Einstein have found another set of transformations which satisfies uniquely the above equations. These formulae of new structure are called the Lorentz-Einstein transformation, and are given by the following equations: $x'=\beta\,(x-vt)$, $y'=y$, $z'=z$, $t'=\beta\,(t-vx/c^2)$; where v is the relative velocity of one system with

*See 'The Velocity of Light', in *Nature*, Aug. 20, 1932, by R. J. Kennedy, and his latest paper (No. 261) in the Bibliography.

respect to the other, c, as usual, represents the velocity of light, and the factor $\beta = 1/\sqrt{(1-v^2/c^2)}$.

The most striking characteristic of these formulae is that if we assume that c, the velocity of light, is '*infinite*', all the expressions containing c^2 would become zero, c^2 entering only in the denominators of fractions. In such a limiting case $\beta = 1/\sqrt{(1-0)} = 1/1 = 1$ and $x' = (x-ut)$, $y' = y$, $z' = z$, $t' = t$ which are the older galilean transformations.

Thus there appears the astonishing fact that all the pre-einsteinian physics and mechanics which involved the structural assumption of the galilean transformation, had a *tacit structural assumption* of the infinite velocity of light. This assumption, *known since 1676* to be false as to facts, remained unnoticed before Einstein.

As $c = 3\times10^{10}$ cm./sec., $c^2 = 9\times10^{20}$ is a very large number, whence the fractions vx/c^2 and v^2/c^2 are very small, and β differs very little from unity.

If we apply the Lorentz-Einstein transformation instead of the older galilean transformation to mechanical problems, the changes are so small that they can hardly be detected by experiments, the terrestrial velocities v^2 or vx being so small in comparison with the square of the velocity of light.

The galilean transformations are *experimentally* shown to be structurally invalid for optical and electrodynamic events. The Lorentz-Einstein transformations satisfy structurally the optical and electrodynamic events, and also apply to the older mechanical problems. We see that the Lorentz-Einstein transformations are *more general*, as they include the galilean transformations as a particular case when we assume $c = \infty$.

In a few instances, where we deal with large velocities, the values of the fractions containing the square of the velocity of light become appreciable and allow experimental testing. As yet all such experiments have verified the Einstein theory.

We should repeat again that the achievement of Einstein was the building of a linguistic system similar in structure to the world, which eliminated a pathological pre-human factor of objectification of terms. Such structural elimination was bound to bring some sanity to our theories; and this fact is independent of experiments in physical laboratories. However, it is gratifying to find that experiments support (1933) the Einstein theory. It was particularly gratifying in the beginning, when physicists and Einstein himself believed that his theory would stand or fall by experiment. Today we see that this theory represents such an enormous general, structural, epistemological, psycho-logical, and methodological *non-elementalistic* advance, that no matter what the experiments show or may show in the future we cannot return to a language of the old, *el*, obviously wrong, structure of the pre-einsteinian days. As usual, the negative results are the important ones. No matter what experiments may show we shall never again accept the silent structural assumption of 'infinite' velocity for light, when we know positively that the velocity is finite. We shall never again treat *terms* of 'matter', 'space', and 'time' as objects—lower order abstractions, when we know that they represent *terms*—higher order abstractions. When

once this is realized, we cannot ascribe 'finiteness' or 'infiniteness', 'definiteness', 'rigidity'. , to *terms*, verbal forms, forms of representation. From this point of view we may consider the Einstein theory as an irreversible gain. If it had achieved only the elimination of various structural prejudices and dogmas, it has done well; and at least this much Einstein has already achieved.

The structural, verbal, cortical quest for invariance in our formulations also becomes apparent. The older mechanics were invariant under the galilean transformation, equations preserved their form in different systems of co-ordinates. In the special theory of relativity the new laws are invariant under the Lorentz-Einstein transformation. In this special, or restricted, theory of relativity only uniform relative motion was taken into account. If we generalize the principle of relativity to *any* kind of relative motion we pass from the restricted to the *general* theory, which demands that the laws of physics should be formulated in a generally invariant form for any arbitrary transformations.

For this structural, cortical reason it is necessary to express all the laws of physics in tensor equations, which satisfy such conditions of general invariance. If this cannot be done, there must be something wrong with our language, as such, and with our verbal laws. We require structural revision of those laws so as to be able to express them in tensor equations. The newtonian law of gravitation and the older form of the law of conservation of energy are perhaps the most remarkable examples. They do not survive such minimal, and yet entirely justified, structural requirements as those of the general theory of Einstein, and therefore they cannot be structurally satisfactory.

We have already seen that the equation $x^2+y^2+z^2=c^2t^2$ or $x^2+y^2+z^2-c^2t^2=0$ represents the equation of the spherical *propagation of light* with the finite velocity c. The discovery that the velocity of light is a universal constant for all observers, and the above equation, led historically to the re-discovery by Einstein of the Lorentz transformation which, as we have seen, has assumed such overwhelming structural importance. The meaning of these facts is worth considering.

In Chapter XVII we analysed briefly the elementalistic language of 'matter', 'space', and 'time', and came to the conclusion that to eliminate objectification we must abandon the semantic disturbance and the use of the term 'is' of identity. Instead, we must use an actional functional language to describe ordered functioning, behaviour, or operations. By necessity we were led to a 'contact' method. We also discovered that in accepting the above structural methods we were compelled to discriminate between different orders of abstraction, since what we see, feel, and experience is *not* what we say about it. We found that on the 'objective' level of our actual activities, (manipulating instruments . ,) which represent the silent *un-speakable* level, we could never find a situation in which the old language of 'matter', 'space', and 'time' could be used without coming violently into conflict with the properly analysed facts. We came to the conclusion that this language was not structurally satisfactory, for *verbally*, 'space', 'time', and 'matter' were supposed to be quite clear-cut and *separate* entities, while in actual experience we *never could find such separated* objective entities. It became obvious that the *structure* of the

old language of 'matter', 'space', and 'time' was *different* from the structure of the outside world as we now know it. We found ourselves in a situation where we had to choose either to keep the old language which, as it differed from them in structure, could never give a coherent account of facts at hand, or else to build up a new language with *structure* similar to that of the outside world, in order to have the possibility of coherent conversation about it.

The invention of such a new language is of course an extremely difficult undertaking. In fact, it requires some genius to invent new, more structurally similar, forms of representation for the old facts. Lorentz, Einstein, and Minkowski prepared and finally produced such a structurally new language. The difficulty was that verbally we had already separated what empirically could not be separated. The problem was to amalgamate somehow the old structurally *elementalistic* language of 'space' *and* 'time' into a *non-elementalistic* language. The key to such an amalgamation is found in the light-wave equation which gives us the structural information about the world, $x^2+y^2+z^2=c^2t^2$.

This equation represents an equality. The left-hand side is expressed in 'spatial' terms only—the *distance* between two points O and P. The right-hand side expresses the 'spatial' length, but in a 'temporal' term. We see that here we have means of translation, and a possibility of amalgamation of two *elementalistic* languages, which were not supposed to be intertranslatable.

The Lorentz-Einstein transformation formulae are $x'=\beta\ (x-vt)$, $y'=y$, $z'=z$, $t'=\beta\ (t-vx/c^2)$ where v is the relative velocity of the two systems of coordinates, c, the constant velocity of light, and $\beta=1/\sqrt{(1-v^2/c^2)}$.

The formulae for x' and t' which typify, on the left-hand side, x' a 'spatial' *length* and t' a 'time', are of particular interest. We see that on the right-hand side of the expressions the value of the 'spatial' x' is given by $\beta\ (x-vt)$ which involves '*time*'. The value of 'time', t' is given by $\beta\ (t-vx/c^2)$, which involves the 'spatial' *length* x. So we see that our amalgamation is complete, and separation impossible. The above formulae express structurally the *simple experimental* fact that 'space' and 'time' cannot be separated. At this point we are not ready to discuss 'matter'. This will be considered further on in this work (see Chapters XL and XLI).

The above formulae have also a very important physical and experimental meaning, as they introduce the 'contact' methods into our language. Our actual measurements of 'space' and 'time' are strictly connected with readings on some instruments, and involve therefore coincidences between pointers and 'simultaneity'. In all instances the finite velocity of propagation of signals must be taken into consideration. When our instrument, or the eye, is affected by signals there is always a delay due to the finite velocity of the propagation of the signals. These delays are part and parcel of our experiment, and so our formulae must contain terms explicitly involving this finite velocity of propagation. This innovation involves not only a most profound structural epistemological and semantic revolution but supplies the very factor that enables us to formulate more structurally satisfactory languages (theories), which Lorentz, Einstein, and Minkowski have produced.

We have been contrasting finite and 'infinite' velocities. Let us say frankly that 'infinite' velocity is a polite way of speaking about blunders of observation. 'Infinite' velocity is meaningless. Velocity is defined as $v = s/t$ and if t is taken as zero or in other words, if one of the fundamental factors in our *definition* is lacking, our *definition ceases to define the term in question*—in this case, velocity. So when the term 'time' is lacking, we have *no velocity*, by definition; so, to speak or speculate about 'infinite' velocity is simply making noises, and not saying anything. The *negative* of this noise; namely, saying that velocity is *not* 'infinite', or in a positive sense, that velocity is 'finite', is on a different verbal footing, although it remains a polite invitation to stop talking non-sense.

It should be noticed carefully that the *general* theory of Einstein is a high structural generalization of the special theory; and that both of them are generalizations of the classical mechanical principle of relativity. It is founded, not on the introduction of any extraordinary structural assumptions, but on the elimination of some unjustified and false-as-to-facts structural assumptions, such as that of the 'infinite' velocity of light.

Both the theory of Einstein, and the theory presented in this work are long overdue. The Einstein theory could have been formulated as soon as we discovered the *finite* velocity of light, in 1676. It should be noticed that this last discovery was also overdue, as it did *not* require experiments to establish the finite velocity of light. It was sufficient to establish the meaningless character of 'infinite' velocity, which on symbolic grounds, could have been accomplished much earlier, and to conclude, that the velocity of light *must* be finite. This example shows the hampering, blocking, semantic effect which different meaningless verbal structures have on us. To express this high and satisfactory structural generalization, Einstein had to select the most general and structurally appropriate language in existence. He chose at some stage of his work the language of $\bar{E}$ and four-dimensional geometries in general and that of the differential geometry and the tensor calculus in particular. In the latest field theory, Einstein and Mayer introduce a new *more general* and very revolutionary mathematical language where vectors and tensors in an n-dimensional spread may have m components.

At present it appears that two other very general mathematical disciplines will be used increasingly in the future. One of them is the *theory of groups;* the other is *analysis situs.* In the latter we study only these characteristics of figures that are unaffected (invariant) by continuous deformation produced without tearing. Two structural points are relevant for us in this connection: namely, that the analysis situs is fundamentally a *differential* and also an *ordinal* discipline, based on asymmetrical relations. In the next chapter, as an illustration of the actional, behaviouristic, functional, operational, differential, contact method a short account will be given of the way Einstein structurally treated 'simultaneity'. The elimination of the old structural dogma about 'simultaneity' resulting from the semantic disturbance of objectification of 'time', is one of the outstanding achievements of Einstein and is historically the beginning of his theory.